Hidden Tails: Certifying Tail-Risk Claims under Selective Labels Supplementary Material

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Scope and reading guide

This supplement collects the full proofs, experimental protocols, and saved results. Appendices A–E share their sources with the paper's integrated appendices. Appendix F develops additional finite-population diagnostics from the experimental archive. The accompanying ZIP supplies code, configurations, numerical outputs, and reproduction checks.

Revision experiments and checks. The finite-cell study separates oracle interval containment from learned-endpoint accuracy. Controlled repayment masks check empirical enlargement and model comparisons against known outcomes. The corrected sequential audit uses the retained plug-in objective, with the historical outcome fit fixed and the predictor retrained after each batch: 20 masks, six policies, and nine recorded rounds, with terminal comparisons paired within mask. The code includes cohort-identity and optimization-equivalence checks. This edition consolidates completed outputs; no new experiment was run.

Question or evidence	Location
Sharpness and compatible population laws	Appendix A
Estimation, statistical calibration, and learning lower bound	Appendix B
Retained-information revelation and minimum-cost allocation	Appendix C
Finite-sample containment and learned-endpoint accuracy	Appendix D.2
Lending cohort support and historical assessment	Appendix D.3
Known-outcome masking and model comparisons	Appendix D.4
Synthetic frontier and corrected repayment audits	Appendices D.5–D.6
Sensitivity calibration and interpretation	Appendix E
Additional mechanism, reference-forecast, and diabetes diagnostics	Appendix F
Commands, environments, and result provenance	`docs/supplement.md` in the ZIP

How to interpret the evidence. Oracle coverage, learned-endpoint error, empirical fold enlargement, and sequential retained width answer different questions. The additional diagnostics in Appendix F use specified finite populations and, where stated, a finite cutoff grid. They do not provide new finite-sample or adaptive confidence guarantees. The diabetes illustration uses all known outcomes to define its strata and observation mechanism; it is not a held-out clinical prediction evaluation.

Using the companion files. Keep this PDF beside `main.pdf` for links to the main text. Shared appendix and bibliography numbering is preserved. The experiment guide maps saved results to commands and separates current runs from archival procedures. Obtain raw borrower records separately using the documented preparation instructions.

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Guide to the appendices

Appendix A gives compatible laws, examples, and identification proofs; Appendix B gives score bias, uniform estimation, calibration, and the learning lower bound; Appendix C gives retained-information revelation, its cost frontier, and computation. Appendix D records scientific protocols and complete experimental results; Appendix E treats sensitivity calibration, extensions, and related approaches. Implementation provenance accompanies the source in `reproducibility/`.

Throughout, $\beta = 1 - \alpha$, $\Gamma < \infty$, and the target population and outcome are fixed. $I_\Gamma(f)$ is identification width; $\widehat{C}_{n,\Gamma}(f)$ is a statistical outer certificate. After revelation, $W_{f,\eta}(\lambda)$ is the gap at a common threshold, whereas $I^{\mathrm{ret}}_{\Gamma,\lambda}(f)$ optimizes the two endpoints separately. Auxiliary results and equations use appendix-local numbering.

A Population Identification and Sensitivity

A.1 Compatible full-data laws and sensitivity bounds

The target outcome and population are fixed across predictors and compatible laws. In a decision problem, $Y = Y(a^\star)$ is the outcome under a specified action (e.g., default under loan issuance), and S records its availability [15–17]. Using an observed outcome requires consistency with that action and outcome definition. Comparing prediction rules holds $a^\star$ and P_X fixed; changing the action, recovering an outcome under an untaken action, or transferring to a new population requires an additional causal or transport argument. The sensitivity model supplies neither exchangeability nor causal identification. Passive revelation recovers an already-realized target outcome, as in record linkage or controlled masking. Thus mature issued loans supply the masking target, whereas repayment under issuance remains a potential outcome for rejected applicants in the historical analysis.

Let $\mathcal{X}$ be standard Borel, and let P_{obs} be the law of $O = (X, S, Y^{\mathrm{obs}})$ with $Y^{\mathrm{obs}} = SY$. Use measurable versions

$$\pi(x) = P_{\mathrm{obs}}(S = 1 \mid X = x), \qquad p_1(x) = P_{\mathrm{obs}}(Y^{\mathrm{obs}} = 1 \mid X = x, S = 1),$$

and losses $z^y_f(x) = \ell(f(x), y) \in [0, 1]$. Assumption 1 gives $\pi \geq \pi_{\min} > 0$, so $\pi\, dP_X$ and P_X have the same null sets and p_1 is identified P_X-almost surely. All unqualified almost-sure statements use P_X. Write $H = \{\pi < 1\}$. For a measurable $q : \mathcal{X} \to [0, 1]$ and its Bernoulli kernel K_q, define

$$\begin{aligned} P^q(dx, \{1\}, dy) &= P_X(dx)\pi(x)K_{p_1}(x, dy), \\ P^q(dx, \{0\}, dy) &= P_X(dx)[1 - \pi(x)]K_q(x, dy). \end{aligned} \tag{A.1}$$

Set $q = p_1$ on H^c, where its value carries zero hidden mass.

Lemma A.1 (Kernel representation of compatible laws)**.** *Every full-data law inducing P_{obs} has the form (A.1), with $q(x) = P(Y = 1 \mid X = x, S = 0)$ on H. Conversely, every measurable $q : \mathcal{X} \to [0, 1]$ in (A.1) induces P_{obs}. With $q = p_1$ on H^c, these kernels identify the compatible laws up to P_X-null sets.*

Proof. Standard Borel disintegration fixes the (X, S) law and selected-label kernel, leaving one binary probability on $S = 0$. Conversely, the two displayed measures sum to one and have the prescribed pushforward under $(x, s, y) \mapsto (x, s, sy)$. Equal full-data laws have equal hidden kernels $(1 - \pi)dP_X$-almost surely, equivalently P_X-almost surely on H; the canonical assignment settles H^c. □

For finite $\Gamma \geq 1$, the odds restriction is equivalent, at interior p_1, to the bounds $\underline{p}_\Gamma = q_-(p_1; \Gamma)$ and $\overline{p}_\Gamma = q_+(p_1; \Gamma)$, where

$$q_-(p; \Gamma) = \frac{p}{p + \Gamma(1 - p)}, \qquad q_+(p; \Gamma) = \frac{\Gamma p}{1 - p + \Gamma p}. \tag{A.2}$$

Their denominators are at least one. Thus $p_1 = 0$ forces $q = 0$ and $p_1 = 1$ forces $q = 1$ at every finite Γ; these substantive boundary restrictions merit scrutiny when apparent certainty comes from

a fitted model. For interior p_1, the endpoints tend to zero and one as $\Gamma \to \infty$. Let $\mathcal{Q}_\Gamma$ contain measurable kernels within these bounds on H, canonically assigned on H^c. Then

$$\mathcal{P}_\Gamma(P_{\text{obs}}) = \{P^q : q \in \mathcal{Q}_\Gamma\}. \tag{A.3}$$

It is nonempty since $\underline{p}_\Gamma \le p_1 \le \overline{p}_\Gamma$; at $\Gamma = 1$ it is the single law satisfying $Y \perp S \mid X$.

For $\beta = 1 - \alpha \in (0, 1)$, the CVaR representation [29] reduces to

$$\rho_{\alpha,P}(f) = \inf_{\eta \in \mathbb{R}} \{\eta + \beta^{-1}\mathbb{E}_P(L_f - \eta)_+\} = \min_{\eta \in [0,1]} \{\eta + \beta^{-1}\mathbb{E}_P(L_f - \eta)_+\}. \tag{A.4}$$

For $\eta < 0$, the objective is $\beta^{-1}\mathbb{E}_P L_f + (1 - \beta^{-1})\eta$ and exceeds its value at zero; for $\eta > 1$, it equals $\eta > 1$. Continuity on $[0, 1]$ gives attainment, including distributions with atoms.

A.2 Observational equivalence and motivating constructions

Theorem A.2 (Observational equivalence with reversed CVaR rankings)**.** *For every $\alpha \in (0, 1)$, there exist an observed law satisfying strict overlap, two predictors f, g with identical selected-loss distributions, and two full-data laws $P^\star, Q^\star$ inducing that observed law such that*

$$\rho_{\alpha,P^\star}(f) < \rho_{\alpha,P^\star}(g), \qquad \rho_{\alpha,Q^\star}(f) > \rho_{\alpha,Q^\star}(g).$$

Proof. Take constant X, $0 < \pi < \min\{1, 2\beta\}$, $S \sim \text{Bernoulli}(\pi)$, and $Y \mid S = 1 \sim \text{Bernoulli}(1/2)$. Under zero-one loss, the constant predictors $f = 0, g = 1$ have the same selected-loss law. Set $Y = 0$ on $S = 0$ under $P^\star$ and $Y = 1$ there under $Q^\star$. Both yield observed masses $\pi/2, \pi/2, 1 - \pi$ on $(S, Y^{\text{obs}}) = (1, 1), (1, 0), (0, 0)$ and satisfy overlap with $\pi_{\min} = \pi$.

For $B \sim \text{Bernoulli}(r)$, the threshold objective is $r/\beta + \eta(1 - r/\beta)$ on $[0, 1]$, hence

$$\text{CVaR}_\alpha(B) = \min\{r/\beta, 1\}. \tag{A.5}$$

Under $P^\star$, the loss probabilities are $\pi/2$ and $1 - \pi/2$, so $\rho_{\alpha,P^\star}(f) = \pi/(2\beta) < \min\{(1 - \pi/2)/\beta, 1\} = \rho_{\alpha,P^\star}(g)$. Under $Q^\star$ they exchange. □

This construction concerns the unrestricted compatible class: its hidden probabilities zero and one lie outside the finite-Γ interval $[1/(1 + \Gamma), \Gamma/(1 + \Gamma)]$ when $p_1 = 1/2$.

Figure 1. For squared loss, $f = 1/5$, $\pi = 1/2$, $p_1 = 1/3$, $\Gamma = 2$, and $\alpha = 1/2$, the loss is $1/25 + (3/5)B$ with $B \sim \text{Bernoulli}(1/6 + q/2)$ and $q \in [1/5, 1/2]$. Translation and positive scaling of (A.4), followed by (A.5), give

$$\begin{aligned} \rho_{1/2,P^q}(f) &= \frac{1}{25} + \frac{6}{5}\left(\frac{1}{6} + \frac{q}{2}\right), \\ \rho_{1/2,P^{1/5}}(f) = \frac{9}{25} = 0.36, &\quad \rho_{1/2,P^{1/2}}(f) = \frac{27}{50} = 0.54. \end{aligned} \tag{A.6}$$

These endpoints describe the risk of one fixed predictor under the same observed law.

A.3 Conditional excess-loss envelopes and common extremal laws

Write $a^y_{f,\eta} = (z^y_f - \eta)_+$, $d_{f,\eta} = a^1_{f,\eta} - a^0_{f,\eta}$, $d_+ = (d_{f,\eta})_+$, and $d_- = (-d_{f,\eta})_+$. The hidden conditional excess loss is $a^0 + qd$, with envelopes

$$\begin{aligned} b^U_{f,\eta} &= a^0_{f,\eta} + \overline{p}_\Gamma d_+ - \underline{p}_\Gamma d_-, \\ b^L_{f,\eta} &= a^0_{f,\eta} + \underline{p}_\Gamma d_+ - \overline{p}_\Gamma d_-. \end{aligned} \tag{A.7}$$

The observable objectives are

$$\begin{aligned} m^\sigma_{f,\eta} &= \mathbb{E}_{\text{obs}}[S a^{Y^{\text{obs}}}_{f,\eta}(X) + (1 - S) b^\sigma_{f,\eta}(X)], \quad \sigma \in \{L, U\}, \\ L_{f,\eta} &= \eta + \beta^{-1} m^L_{f,\eta}, \qquad U_{f,\eta} = \eta + \beta^{-1} m^U_{f,\eta}. \end{aligned} \tag{A.8}$$

Lemma A.3 (Common extremal kernels and threshold regularity)**.** *On H, define*

$$\begin{aligned} q_f^U(x) &= \overline{p}_\Gamma(x)\mathbf{1}\{z_f^1(x) \geq z_f^0(x)\} + \underline{p}_\Gamma(x)\mathbf{1}\{z_f^1(x) < z_f^0(x)\}, \\ q_f^L(x) &= \underline{p}_\Gamma(x)\mathbf{1}\{z_f^1(x) \geq z_f^0(x)\} + \overline{p}_\Gamma(x)\mathbf{1}\{z_f^1(x) < z_f^0(x)\}, \end{aligned} \tag{A.9}$$

and set both kernels to p_1 on H^c. These kernels belong to $\mathcal{Q}_\Gamma$. For every $q \in \mathcal{Q}_\Gamma$ and every $\eta \in [0,1]$,

$$m_{f,\eta}^L \leq \mathbb{E}_{P^q}[(L_f - \eta)_+] \leq m_{f,\eta}^U, \qquad \mathbb{E}_{P^{q_f^\sigma}}[(L_f - \eta)_+] = m_{f,\eta}^\sigma. \tag{A.10}$$

The objectives $L_{f,\eta}, U_{f,\eta}$ are convex and $(1+\beta^{-1})$-Lipschitz in η, and attain their minima on $[0,1]$.

Proof. The kernels are measurable and satisfy the bounds. Since $z \mapsto (z-\eta)_+$ is nondecreasing, thresholding preserves the ordering of z_f^1, z_f^0; it can create a tie but cannot reverse the sign. Thus the same kernels extremize the affine expression $a^0 + qd$ at every threshold, including ties. Disintegration gives

$$\begin{aligned} \mathbb{E}_{P^q}(L_f - \eta)_+ = \mathbb{E}_{\mathrm{obs}}[S a_{f,\eta}^{Y^{\mathrm{obs}}}(X)] \\ + \mathbb{E}_{P_X}[(1-\pi)(a_{f,\eta}^0 + q d_{f,\eta})], \end{aligned} \tag{A.11}$$

proving (A.10), with zero contribution from H^c. Each endpoint objective is therefore the CVaR objective of one fixed law. Convexity and the Lipschitz bound follow from the convex, 1-Lipschitz hinge; compactness gives attainment. □

A.4 Sharpness, endpoint attainment, and interval filling

Proof of Theorem 1. Put $G_f(q,\eta) = \eta + \beta^{-1}\mathbb{E}_{P^q}(L_f - \eta)_+$. Lemma A.3 gives $L_{f,\eta} = G_f(q_f^L, \eta) \leq G_f(q,\eta) \leq G_f(q_f^U, \eta) = U_{f,\eta}$ simultaneously for all thresholds. Minimizing these inequalities and choosing the respective common kernels proves

$$\underline{\rho}_{\alpha,\Gamma}(f) = \min_{\eta\in[0,1]} L_{f,\eta}, \qquad \overline{\rho}_{\alpha,\Gamma}(f) = \min_{\eta\in[0,1]} U_{f,\eta}. \tag{A.12}$$

Both threshold minima and both optimized endpoints are attained. In particular, the upper equality uses one kernel attaining the supremum at every threshold, without an abstract minimax interchange.

For functions with finite infima on a common domain,

$$\left|\inf_\eta F(\eta) - \inf_\eta G(\eta)\right| \leq \sup_\eta |F(\eta) - G(\eta)|, \tag{A.13}$$

because each function is bounded below by the other minus their uniform distance. Equation (A.11) and $|d_{f,\eta}| \leq 1$ therefore imply

$$|\rho_{\alpha,P^q}(f) - \rho_{\alpha,P^{\widetilde{q}}}(f)| \leq \sup_\eta |G_f(q,\eta) - G_f(\widetilde{q},\eta)| \leq \beta^{-1}\mathbb{E}[(1-\pi)|q - \widetilde{q}|].$$

The mixture $q_t = (1-t)q_f^L + t q_f^U$, $t \in [0,1]$, remains measurable and compatible, and satisfies

$$|\rho_{\alpha,P^{q_t}}(f) - \rho_{\alpha,P^{q_s}}(f)| \leq \frac{|t-s|}{\beta}\mathbb{E}[(1-\pi)|q_f^U - q_f^L|] \leq \frac{|t-s|}{\beta}. \tag{A.14}$$

This continuous path joins the two attained endpoints. The intermediate value theorem, including the singleton case, yields

$$\{\rho_{\alpha,P}(f) : P \in \mathcal{P}_\Gamma(P_{\mathrm{obs}})\} = [\underline{\rho}_{\alpha,\Gamma}(f), \overline{\rho}_{\alpha,\Gamma}(f)]. \tag{A.15}$$

□

A.5 Sensitivity analysis and population decision criteria

Proposition A.4 (Nested classes and continuous endpoint curves)**.** *If $1 \le \Gamma_1 \le \Gamma_2 < \infty$, then*

$$\mathcal{P}_{\Gamma_1}(P_{\mathrm{obs}}) \subseteq \mathcal{P}_{\Gamma_2}(P_{\mathrm{obs}}).$$

For each f, the lower endpoint is nonincreasing in Γ, the upper endpoint is nondecreasing, and the identification width $I_\Gamma(f) = \overline{\rho}_{\alpha,\Gamma}(f) - \underline{\rho}_{\alpha,\Gamma}(f)$ is nondecreasing. The interval is a singleton at $\Gamma = 1$. Moreover, for $\sigma \in \{L, U\}$, writing $\rho^L_{\alpha,\Gamma} = \underline{\rho}_{\alpha,\Gamma}$ and $\rho^U_{\alpha,\Gamma} = \overline{\rho}_{\alpha,\Gamma}$,

$$|\rho^\sigma_{\alpha,\Gamma_2}(f) - \rho^\sigma_{\alpha,\Gamma_1}(f)| \le \frac{\mathbb{E}_{P_X}(1-\pi)}{4\beta}|\Gamma_2 - \Gamma_1|. \tag{A.16}$$

In particular, both endpoint curves are continuous at every finite sensitivity level.

Proof. For $p \in [0,1]$, $\partial_\Gamma q_- = -p(1-p)/\{p+\Gamma(1-p)\}^2 \in [-1/4, 0]$ and $\partial_\Gamma q_+ = p(1-p)/\{1-p+\Gamma p\}^2 \in [0, 1/4]$. Their signs prove nesting and endpoint monotonicity; at one both maps equal p. Their magnitude bounds and the Γ-independent loss ordering in (A.9) bound the difference of the extremal kernels by $|\Gamma_2 - \Gamma_1|/4$. Apply (A.11) and (A.13) to obtain (A.16). □

Corollary A.5 (Population risk budgets, comparisons, and selection)**.** *For any risk budget r,*

$$\rho_{\alpha,P}(f) \le r \quad \textit{for every } P \in \mathcal{P}_\Gamma(P_{\mathrm{obs}}) \quad \Longleftrightarrow \quad \overline{\rho}_{\alpha,\Gamma}(f) \le r. \tag{A.17}$$

For two predictors, the sufficient separation condition

$$\overline{\rho}_{\alpha,\Gamma}(f) < \underline{\rho}_{\alpha,\Gamma}(g) \tag{A.18}$$

implies $\rho_{\alpha,P}(f) < \rho_{\alpha,P}(g)$ under every compatible law. If $\overline{\rho}_{\alpha,\Gamma}(f_\varepsilon) \le \inf_{g\in\mathcal{F}} \overline{\rho}_{\alpha,\Gamma}(g) + \varepsilon$, then every such law satisfies $\rho_{\alpha,P}(f_\varepsilon) \le \inf_{g\in\mathcal{F}} \overline{\rho}_{\alpha,\Gamma}(g) + \varepsilon$. A nonempty finite class permits $\varepsilon = 0$.

Proof. Use the attained upper endpoint for (A.17), the chain $\rho_P(f) \le \overline{\rho}(f) < \underline{\rho}(g) \le \rho_P(g)$ for comparison, and $\rho_P(f_\varepsilon) \le \overline{\rho}(f_\varepsilon)$ for selection. □

If $\underline{\rho}(f) > r$, every law violates the budget; if $\underline{\rho}(f) \le r < \overline{\rho}(f)$, the two extremal laws respectively satisfy and violate it. Thus sharpness covers all budget and equality cases.

Common-law comparisons. Define $D_{f,g}(\Gamma) = \overline{\rho}_{\alpha,\Gamma}(f) - \underline{\rho}_{\alpha,\Gamma}(g)$. Marginal extrema can use different hidden kernels, whereas the direct contrast is

$$\Delta_\Gamma(f,g) = \sup_{P\in\mathcal{P}_\Gamma(P_{\mathrm{obs}})} \{\rho_{\alpha,P}(f) - \rho_{\alpha,P}(g)\} \le D_{f,g}(\Gamma). \tag{A.19}$$

Consequently, $D_{f,g} < 0$ proves common-law dominance, but overlapping marginal intervals leave its sign unresolved by this criterion. Determining Δ_Γ requires joint optimization under one hidden kernel. The uniform event of Appendix B.4 makes the marginal comparison and selection rules valid simultaneously over the stipulated predictor class using the same observations.

First breakdown of an interval-certified comparison. Set $\Gamma_{\mathrm{break}}(f,g) = \inf\{\Gamma \ge 1 : D_{f,g}(\Gamma) \ge 0\}$, with $\inf \varnothing = \infty$. The gap is continuous and nondecreasing. If $D_{f,g}(1) \ge 0$, breakdown is one and no strict interval certificate holds. If $D_{f,g}(1) < 0$, breakdown exceeds one; when finite, continuity forces $D_{f,g}(\Gamma_{\mathrm{break}}) = 0$ and strict separation holds exactly on $[1, \Gamma_{\mathrm{break}})$. A nonzero gap at that boundary would persist across it, contradicting first failure. Infinite breakdown means separation at every finite level. Differences between two nondecreasing upper endpoints need not be monotone, so the upper-endpoint minimizer may change independently of these pairwise breakdown values.

B Statistical Certificates and Learning Complexity

This appendix proves the statistical results for a fixed tail level $\alpha \in (0,1)$ and a fixed finite sensitivity level $\Gamma \geq 1$. Write $\beta = 1-\alpha$, $p = p_1$, and $P = P_{\mathrm{obs}}$; thus Pg denotes integration of an observed-data function g against the common observed law. For brevity, set

$$\rho^L(f) = \underline{\rho}_{\alpha,\Gamma}(f), \qquad \rho^U(f) = \overline{\rho}_{\alpha,\Gamma}(f), \qquad I_\Gamma(f) = \rho^U(f) - \rho^L(f).$$

The population characterization in Appendix A gives $\rho^\sigma(f) = \inf_{\eta\in[0,1]}\{\eta + \beta^{-1} m^\sigma_{f,\eta}\}$ for $\sigma \in \{L,U\}$. All L_2 norms below are with respect to P_X.

For uniform empirical-process statements, assume the paired maps $x \mapsto (\ell(f(x),0), \ell(f(x),1))$, $f \in \mathcal{F}$, form a pointwise-separable class of measurable functions with the envelope supplied by Assumption 1. A countable pointwise-dense subclass and rational thresholds including $0, 1$ determine the suprema below, since the score transformations are continuous. Measurable, pointwise-separable versions of the fitted score classes suffice as well.

B.1 Score regularity and exact nuisance-bias expansion

Recall $a^y_{f,\eta}(x) = (\ell(f(x),y) - \eta)_+$ and $d_{f,\eta}(x) = a^1_{f,\eta}(x) - a^0_{f,\eta}(x)$. For a real number d, write $d_+ = \max(d,0)$ and $d_- = \max(-d,0)$. Define

$$H^U_p(d) = q_+(p)d_+ - q_-(p)d_-, \qquad H^L_p(d) = q_-(p)d_+ - q_+(p)d_-, \tag{B.1}$$

$$G^U_p(d) = q'_+(p)d_+ - q'_-(p)d_-, \qquad G^L_p(d) = q'_-(p)d_+ - q'_+(p)d_-. \tag{B.2}$$

Consequently, $b^\sigma_{f,\eta}(x;p) = a^0_{f,\eta}(x) + H^\sigma_p(d_{f,\eta}(x))$ and $\dot{b}^\sigma_{f,\eta}(x;p) = G^\sigma_p(d_{f,\eta}(x))$.

Lemma B.1 (Regularity of the endpoint maps)**.** *For $p \in [0,1]$, the derivatives of the sensitivity endpoints satisfy*

$$q'_+(p) = \frac{\Gamma}{\{1+(\Gamma-1)p\}^2}, \qquad q'_-(p) = \frac{\Gamma}{\{\Gamma-(\Gamma-1)p\}^2}, \tag{B.3}$$

$$q''_+(p) = -\frac{2\Gamma(\Gamma-1)}{\{1+(\Gamma-1)p\}^3}, \qquad q''_-(p) = \frac{2\Gamma(\Gamma-1)}{\{\Gamma-(\Gamma-1)p\}^3}. \tag{B.4}$$

In particular, $\Gamma^{-1} \leq q'_\pm(p) \leq \Gamma$ and $|q''_\pm(p)| \leq 2\Gamma(\Gamma-1)$. Uniformly in (f,η,x,p,σ),

$$\begin{gathered} 0 \leq b^\sigma_{f,\eta}(x;p) \leq 1, \qquad |\dot{b}^\sigma_{f,\eta}(x;p)| \leq \Gamma, \\ |\partial_p^2 b^\sigma_{f,\eta}(x;p)| \leq 2\Gamma(\Gamma-1). \end{gathered} \tag{B.5}$$

Moreover, H^σ_p is 1-Lipschitz and G^σ_p is Γ-Lipschitz on the real line; both vanish at zero.

Proof. Differentiating $q_+(p) = \Gamma p/\{1+(\Gamma-1)p\}$ and $q_-(p) = p/\{\Gamma-(\Gamma-1)p\}$ gives (B.3)–(B.4). The denominators before taking powers lie in $[1,\Gamma]$, giving the derivative bounds. The value $a^0 + H^\sigma_p(a^1 - a^0)$ is a convex combination of $a^0, a^1 \in [0,1]$; its derivatives are bounded by $\Gamma|a^1-a^0|$ and $2\Gamma(\Gamma-1)|a^1-a^0|$, respectively. This proves (B.5). On either half-line the slopes of H^σ_p and G^σ_p lie in $[0,1]$ and $[\Gamma^{-1},\Gamma]$, respectively. Both maps are continuous at zero, so these slope bounds also prove the Lipschitz bounds across loss ties. □

For candidate nuisance functions $p_c : \mathcal{X} \to [0,1]$ and $\pi_c : \mathcal{X} \to [\pi_{\min}/2, 1]$, the score is

$$\begin{aligned} \phi^\sigma_{f,\eta}(O;p_c,\pi_c) = {} & S a^{Y^{\mathrm{obs}}}_{f,\eta}(X) + (1-S) b^\sigma_{f,\eta}(X;p_c(X)) \\ & + S\frac{1-\pi_c(X)}{\pi_c(X)} \dot{b}^\sigma_{f,\eta}(X;p_c(X))\{Y^{\mathrm{obs}} - p_c(X)\}. \end{aligned} \tag{B.6}$$

The sum of its first two terms belongs to $[0,1]$, whereas its correction has absolute value at most $2\Gamma/\pi_{\min}$. Therefore

$$\begin{gathered} -\frac{2\Gamma}{\pi_{\min}} \leq \phi^\sigma_{f,\eta}(O;p_c,\pi_c) \leq 1 + \frac{2\Gamma}{\pi_{\min}}, \\ |\phi^\sigma_{f,\eta}(O;p_c,\pi_c)| \leq B := 1 + \frac{2\Gamma}{\pi_{\min}}. \end{gathered} \tag{B.7}$$

The score can be negative; the concentration argument below explicitly allows signed classes.

Lemma B.2 (Exact conditional nuisance bias)**.** *Suppress (f, η, σ, X) inside b and its derivatives. With $e = p_c - p$, the conditional difference between the score expectation and the target excess-loss integrand is exactly*

$$(1-\pi)\{b(p_c) - b(p) - \dot{b}(p_c)e\} + \frac{\pi_c - \pi}{\pi_c}\dot{b}(p_c)e. \tag{B.8}$$

Taking P_X expectations of (B.8) *gives $P\phi^{\sigma}_{f,\eta}(\cdot; p_c, \pi_c) - m^{\sigma}_{f,\eta}$, and*

$$\begin{aligned}&\left|P\phi^{\sigma}_{f,\eta}(\cdot; p_c, \pi_c) - m^{\sigma}_{f,\eta}\right| \\ &\quad\le \Gamma(\Gamma-1)\|p_c - p\|_2^2 + \frac{2\Gamma}{\pi_{\min}}\|p_c - p\|_2\|\pi_c - \pi\|_2.\end{aligned} \tag{B.9}$$

Proof. Conditional on X, the observed excess-loss contribution has mean $\pi\{(1-p)a^0 + pa^1\}$ and $\mathbb{E}[S(Y^{\mathrm{obs}} - p_c) \mid X] = \pi(p - p_c)$. Subtracting the target integrand $\pi\{(1-p)a^0 + pa^1\} + (1-\pi)b(p)$ leaves

$$(1-\pi)\{b(p_c) - b(p)\} + \frac{\pi(1-\pi_c)}{\pi_c}\dot{b}(p_c)(p - p_c).$$

The identity $\pi(1-\pi_c)/\pi_c = (1-\pi) + (\pi - \pi_c)/\pi_c$ reduces this expression to (B.8). Taylor's theorem, expanded about p_c along the segment joining p_c and p, and Lemma B.1 give

$$|b(p_c) - b(p) - \dot{b}(p_c)(p_c - p)| \le \Gamma(\Gamma-1)(p_c - p)^2.$$

Use $0 \le 1 - \pi \le 1$ for the first term, and $|\dot{b}(p_c)/\pi_c| \le 2\Gamma/\pi_{\min}$ followed by Cauchy–Schwarz for the second term. This yields (B.9) uniformly over all score indices. □

The score is unbiased at the true nuisances and has $O(t^2)$ bias along admissible nuisance paths with L_2 perturbations of order t. The sign of $d_{f,\eta}(x)$ is independent of the nuisance argument, so endpoint selection adds no remainder. At $\Gamma = 1$ the curvature term vanishes; for $\Gamma > 1$, knowing π alone generally leaves quadratic error in p. Thus the result is an orthogonal expansion, not generic double robustness. Its overlap premise and clipping ranges are substantive: clipping cannot supply labels or identification where overlap fails.

B.2 Cross-fitting and uniform empirical-process control

Let $I_1, \dots, I_K$ be a deterministic partition of the observation indices, with $n_k = |I_k| \ge 1$, $n_{\min} = \min_k n_k$, and $w_k = n_k/n$. A random partition independent of the observations is also allowed by conditioning on that partition. Write D_{-k} for the training observations outside I_k and $P_{n,k}$ for the empirical measure on I_k. The clipped fits $\widehat{p}^{(-k)}, \widehat{\pi}^{(-k)}$ are measurable functions of D_{-k}. Any independent fitting randomization can be included in this conditioning. Define

$$\phi^{\sigma,(k)}_{f,\eta} = \phi^{\sigma}_{f,\eta}(\cdot; \widehat{p}^{(-k)}, \widehat{\pi}^{(-k)}), \qquad \Phi^{\sigma}_k = \{\phi^{\sigma,(k)}_{f,\eta} : f \in \mathcal{F},\ \eta \in [0,1]\}.$$

The empirical Rademacher complexities used here are

$$\begin{aligned}\widehat{\mathfrak{R}}^{\sigma,k}_{\mathrm{abs}} &= \mathbb{E}_{\varepsilon}\left[\sup_{f,\eta}\left|\frac{1}{n_k}\sum_{i\in I_k}\varepsilon_i\phi^{\sigma,(k)}_{f,\eta}(O_i)\right| \,\middle|\, D_n\right], \\ \widehat{\mathfrak{R}}^{\sigma}_{n,\mathrm{abs}} &= \sum_{k=1}^{K} w_k \widehat{\mathfrak{R}}^{\sigma,k}_{\mathrm{abs}}.\end{aligned} \tag{B.10}$$

Here D_n includes the observations, partition, and fitting seeds. The expectation averages only independent uniform signs in $\{-1, 1\}$; the fitted functions and evaluation observations are fixed.

Lemma B.3 (Empirical-complexity bound for a signed class)**.** *Let $Z_1, \dots, Z_m$ be independent with law Q and empirical measure Q_m, and let $\mathcal{H}$ be a fixed pointwise-separable class with $|h| \le B$. For $0 < \xi < 1$, with probability at least $1 - \xi$,*

$$\sup_{h\in\mathcal{H}}|(Q_m - Q)h| \le 2\widehat{\mathfrak{R}}_{m,\mathrm{abs}}(\mathcal{H}) + 6B\sqrt{\frac{\log(2/\xi)}{2m}}, \tag{B.11}$$

where the empirical complexity uses the absolute supremum as in (B.10)*.*

Proof. Put $Z = \sup_h |(Q_m - Q)h|$ and $R = \widehat{\mathfrak{R}}_{m,\mathrm{abs}}(\mathcal{H})$. Symmetrization gives $\mathbb{E}Z \le 2\mathbb{E}R$ [3, 33]. Changing one observation changes each of Z, R by at most $2B/m$. With $t = 2B\sqrt{\log(2/\xi)/(2m)}$, McDiarmid's inequality bounds each of $\mathbb{P}(Z > \mathbb{E}Z + t)$ and $\mathbb{P}(\mathbb{E}R > R + t)$ by $\xi/2$. Thus $Z \le 2R + 3t$ with probability at least $1 - \xi$, proving (B.11) for signed classes with $|h| \le B$. □

Conditional on D_{-k}, the observations in I_k are independent with law P, and Φ_k^σ is fixed with envelope B. Apply Lemma B.3 with $m = n_k$ and $\xi = \delta/(2K)$. Its conditional failure probability is at most $\delta/(2K)$, so the same bound holds unconditionally. A union bound over the $2K$ fold–endpoint pairs yields an event $\mathcal{E}_{\mathrm{emp}}$ of probability at least $1 - \delta$ on which

$$\begin{aligned}\sup_{f,\eta}\left|\sum_{k=1}^{K} w_k (P_{n,k} - P)\phi_{f,\eta}^{\sigma,(k)}\right| &\le 2\widehat{\mathfrak{R}}_{n,\mathrm{abs}}^{\sigma} + 6B\sum_{k=1}^{K} w_k\sqrt{\frac{\log(4K/\delta)}{2n_k}} \\ &\le 2\widehat{\mathfrak{R}}_{n,\mathrm{abs}}^{\sigma} + 6B\sqrt{\frac{\log(4K/\delta)}{2n_{\min}}} \end{aligned} \tag{B.12}$$

for both endpoints. Here $P\phi_{f,\eta}^{\sigma,(k)}$ integrates the fitted function against a fresh observation. The training sets overlap; no independence between folds or between their success events is used. In particular, the proof never conditions simultaneously on all fitted nuisances, which could destroy the needed held-out independence.

B.3 Uniform endpoint estimation and complexity rates

Define the cross-fitted empirical objectives and their infima by

$$\begin{aligned}\widehat{Q}_f^{\sigma}(\eta) &= \eta + \frac{1}{\beta}\sum_{k=1}^{K} w_k P_{n,k}\phi_{f,\eta}^{\sigma,(k)}, \\ \widehat{\rho}^{\sigma}(f) &= \inf_{\eta\in[0,1]} \widehat{Q}_f^{\sigma}(\eta).\end{aligned} \tag{B.13}$$

Proof of Theorem 2. Under the theorem's assumptions and the measurability convention above, let $\mathcal{E}_\nu$ be the simultaneous nuisance event

$$\max_k \|\widehat{p}^{(-k)} - p\|_2 \le r_p, \qquad \max_k \|\widehat{\pi}^{(-k)} - \pi\|_2 \le r_\pi, \tag{B.14}$$

with probability at least $1 - \delta_\nu$. For $\delta \in (0,1)$ and $\delta_\nu \in [0, 1-\delta)$, the explicit calibrated radius is

$$\begin{aligned}r_n(\delta) = \frac{1}{\beta}\Bigg[2\max_{\sigma\in\{L,U\}}\widehat{\mathfrak{R}}_{n,\mathrm{abs}}^{\sigma} + 6B\sqrt{\frac{\log(4K/\delta)}{2n_{\min}}} \\ + \Gamma(\Gamma-1)r_p^2 + \frac{2\Gamma}{\pi_{\min}} r_p r_\pi\Bigg].\end{aligned} \tag{B.15}$$

On $\mathcal{E}_\nu$, Lemma B.2, applied to each realized pair of clipped fits, implies

$$\sup_{f,\eta,\sigma}\left|\sum_{k=1}^{K} w_k P\phi_{f,\eta}^{\sigma,(k)} - m_{f,\eta}^{\sigma}\right| \le \Gamma(\Gamma-1)r_p^2 + \frac{2\Gamma}{\pi_{\min}} r_p r_\pi.$$

The intersection $\mathcal{E} = \mathcal{E}_\nu \cap \mathcal{E}_{\mathrm{emp}}$ has probability at least $1 - \delta - \delta_\nu$ by the union bound. On this event, (B.12) and the last display give

$$\sup_{f,\eta,\sigma}\left|\sum_{k=1}^{K} w_k P_{n,k}\phi_{f,\eta}^{\sigma,(k)} - m_{f,\eta}^{\sigma}\right| \le \beta r_n(\delta).$$

Applying the infimum stability inequality (A.13) to $\widehat{Q}_f^{\sigma}$ and its population counterpart yields, on the same event of probability at least $1 - \delta - \delta_\nu$,

$$\sup_{f\in\mathcal{F}}\max_{\sigma\in\{L,U\}} |\widehat{\rho}^{\sigma}(f) - \rho^{\sigma}(f)| \le r_n(\delta). \tag{B.16}$$

This requires neither convexity of the signed empirical objective nor a unique threshold or loss density there. □

For a total confidence level $1-\delta_{\mathrm{tot}}$, allocate $\delta+\delta_\nu \le \delta_{\mathrm{tot}}$. The empirical complexity in (B.15) is an expectation over all Rademacher signs, conditional on the realized data. A Monte Carlo estimate is not automatically an upper bound for this expectation. Any random complexity or nuisance-error bound used instead must hold simultaneously on a specified validity event, whose failure probability is included in the confidence allocation.

We next bound the score complexity. For a sample $x_1,\dots,x_m$, let $\widehat{\mathfrak{R}}_{m,\mathrm{abs}}(\mathcal{V})$ denote its absolute empirical complexity. Define

$$\mathcal{L}_y=\{x\mapsto \ell(f(x),y): f\in\mathcal{F}\},\qquad \mathcal{A}_y=\{x\mapsto(\ell(f(x),y)-\eta)_+ : f\in\mathcal{F},\ \eta\in[0,1]\}.$$

Define the threshold-augmented loss-class complexity by

$$\mathfrak{C}_n=\sum_{k=1}^{K} w_k\left\{\widehat{\mathfrak{R}}_{n_k,\mathrm{abs}}(\mathcal{L}_0)+\widehat{\mathfrak{R}}_{n_k,\mathrm{abs}}(\mathcal{L}_1)+n_k^{-1/2}\right\}, \tag{B.17}$$

where each term uses the covariates in its evaluation fold. For fixed fitted nuisances and $o=(x,s,y^{\mathrm{obs}})$, put $p_c=\widehat{p}^{(-k)}(x)$ and $w=(1-\widehat{\pi}^{(-k)}(x))/\widehat{\pi}^{(-k)}(x)$. The score is $\Psi_o^\sigma(a^0_{f,\eta}(x),a^1_{f,\eta}(x))$, where

$$\begin{aligned}\Psi_o^\sigma(u_0,u_1)=\ & s\{(1-y^{\mathrm{obs}})u_0+y^{\mathrm{obs}}u_1\}\\ &+(1-s)\{u_0+H^\sigma_{p_c}(u_1-u_0)\}\\ &+swG^\sigma_{p_c}(u_1-u_0)(y^{\mathrm{obs}}-p_c).\end{aligned} \tag{B.18}$$

It vanishes at $(0,0)$. Lemma B.1, $w\le 2/\pi_{\min}$, and $|\Delta u_1-\Delta u_0|\le 2\|\Delta u\|_\infty$ show that it is $L_*=3+4\Gamma/\pi_{\min}$-Lipschitz in the maximum norm, and hence also in Euclidean norm.

The vector-contraction inequality [24] therefore gives, for a universal constant C,

$$\begin{aligned}\widehat{\mathfrak{R}}^{\sigma,k}_{\mathrm{abs}}\le CL_*\Big\{&\widehat{\mathfrak{R}}_{n_k,\mathrm{abs}}(\mathcal{L}_0)+\widehat{\mathfrak{R}}_{n_k,\mathrm{abs}}(\mathcal{L}_1)\\ &+n_k^{-1/2}\Big\}.\end{aligned} \tag{B.19}$$

To handle the absolute supremum, use that the score class contains zero at $\eta=1$: the absolute supremum is at most the sum of the positive and negative suprema, whose expectations agree by sign symmetry. Vector contraction on the positive supremum, followed by separating the two coordinate suprema, bounds it by a universal multiple of $L_*\sum_y\widehat{\mathfrak{R}}_{n_k,\mathrm{abs}}(\mathcal{A}_y)$. Apply the same argument to the $\sqrt{2}$-Lipschitz hinge map $(z,t)\mapsto(z-t)_+$, adjoining zero to its input class if needed. Its coordinate complexities are those of $\mathcal{L}_y$ and the constant threshold class; the latter is $m^{-1}\mathbb{E}_\varepsilon|\sum_i\varepsilon_i|\le m^{-1/2}$. This proves (B.19). All contraction maps are continuous endpoint values or derivatives, including at ties; no discontinuous selector class is introduced.

If $\mathcal{L}_y$ has VC-subgraph dimension at most $d\ge 1$, then $\log N(\epsilon,\mathcal{L}_y,L_2(Q))\le C_0 d\log(C_1/\epsilon)$ for $0<\epsilon<1$ and every probability measure Q. After adjoining zero and the negative class, the standard entropy-to-complexity bound [3, 33] gives, uniformly over samples,

$$\widehat{\mathfrak{R}}_{m,\mathrm{abs}}(\mathcal{L}_y)\le C'\sqrt{\frac{d+1}{m}}. \tag{B.20}$$

It therefore applies to the fixed fitted nuisances in each conditional argument.

For fixed K and balanced folds, $n_{\min}\ge c_K n$ for a positive constant c_K once all folds are nonempty. Equations (B.19)–(B.20) give

$$\begin{aligned}&\frac{1}{\beta}\left[2\max_\sigma\widehat{\mathfrak{R}}^\sigma_{n,\mathrm{abs}}+6B\sqrt{\frac{\log(4K/\delta)}{2n_{\min}}}\right]\\ &\qquad\le\frac{C_{\Gamma,\pi_{\min},K}}{\beta}\sqrt{\frac{d+\log(1/\delta)}{n}}.\end{aligned} \tag{B.21}$$

For fixed $\Gamma,\pi_{\min},\beta$, nuisance rates $o(n^{-1/4})$ make the nuisance contribution in (B.15) $o(n^{-1/2})$. Joint high-probability nuisance bounds remain necessary; cross-fitting does not establish them. All uniformity is over f,η at fixed α,Γ.

Example B.4 (Conditional sparse-nuisance specialization)**.** Suppose sparse logistic nuisance fits in D covariates satisfy simultaneous $L_2(P_X)$ error bounds of orders

$$r_p \lesssim \sqrt{\frac{s_p \log(eD)}{n_{\mathrm{sel}}}}, \qquad r_\pi \lesssim \sqrt{\frac{s_\pi \log(eD)}{n_{\mathrm{tr}}}},$$

where s_p, s_π are sparsities, n_{tr} is the smallest training-fold size, and $n_{\mathrm{sel}} > 0$ is a simultaneous lower bound for the number of revealed training labels in every fold. Substitution in the proved remainder gives, up to the constants in these assumed rates,

$$\frac{1}{\beta}\left\{\Gamma(\Gamma-1)\frac{s_p \log(eD)}{n_{\mathrm{sel}}} + \frac{2\Gamma}{\pi_{\min}}\frac{\sqrt{s_p s_\pi}\log(eD)}{\sqrt{n_{\mathrm{sel}} n_{\mathrm{tr}}}}\right\}.$$

For a fixed number of balanced folds with nonempty training sets, $n_{\mathrm{tr}} \asymp n$. These substitutions require the stated model, design, tuning, and confidence conditions. A selected-population bound transfers to P_X because, for every measurable e,

$$\|e\|^2_{L_2(P_X)} \le \frac{P(S=1)}{\pi_{\min}}\|e\|^2_{L_2(P_{X\mid S=1})}. \tag{B.22}$$

Indeed, $\mathbb{E}[\pi(X)e(X)^2] = P(S=1)\mathbb{E}[e(X)^2 \mid S=1]$ and $\pi(X) \ge \pi_{\min}$. Learning p from revealed outcomes must therefore control target-population error.

Proposition B.5 (Finite threshold search for the fitted objectives)**.** *For a fixed f and realized nuisance fits, each objective in* (B.13) *attains its minimum on*

$$T_n(f) = \{0,1\} \cup \{\ell(f(X_i), y) : 1 \le i \le n,\ y \in \{0,1\}\}.$$

Proof. Thresholding preserves the label-loss ordering at each X_i. Thus $H^\sigma_{p_c}(d_{f,\eta})$ and $G^\sigma_{p_c}(d_{f,\eta})$ are fixed coefficients times $d_{f,\eta}$ throughout $[0,1]$; at $d_{f,\eta} = 0$ both representations vanish. Each score is a fixed linear combination of the two label-specific hinges, making the objective continuous and affine between successive knots of $T_n(f)$. Each such interval has an endpoint minimizer, even if the objective is constant there. Signed correction coefficients may prevent convexity but preserve this result. □

An exact implementation can sort the at most $2n+2$ breakpoints once per predictor. Write the fitted score as $c^\sigma_{i0}(\ell(f(X_i),0)-\eta)_+ + c^\sigma_{i1}(\ell(f(X_i),1)-\eta)_+$, with coefficients fixed across η. Starting from the objective at zero, sweep the sorted breakpoints: the objective is updated by its current slope times the distance to the next breakpoint, and the slope increases by $c^\sigma_{iy}/(n\beta)$ when the corresponding hinge vanishes. Combine all coefficients at a tied breakpoint and initialize the right slope using only losses above zero. Checking all knot values gives both global minima in $O(n \log n)$ time and $O(n)$ storage, after the fitted probabilities and label-specific losses have been evaluated.

B.4 Outer certificates, comparisons, and predictor selection

Proof of the decision guarantees in Section 4.2. Work on the joint event (B.16), of probability at least $1-\delta-\delta_\nu$, and use the certificate (8). Since $0 \le \rho^L(f) \le \rho^U(f) \le 1$, clipping preserves containment of the sharp interval and hence every compatible CVaR. Containment gives the lower width bound; clipping decreases width, while

$$\widehat{\rho}^U(f) - \widehat{\rho}^L(f) + 2r_n \le \rho^U(f) - \rho^L(f) + 4r_n,$$

proving the upper bound. If $\widehat{\rho}^U(f) + r_n < \widehat{\rho}^L(g) - r_n$, then for every compatible law,

$$\begin{aligned}\rho_{\alpha,P^\star}(f) &\le \rho^U(f) \le \widehat{\rho}^U(f) + r_n \\ &< \widehat{\rho}^L(g) - r_n \le \rho^L(g) \le \rho_{\alpha,P^\star}(g).\end{aligned}$$

Finally, any measurable $\varepsilon_{\mathrm{opt}}$-approximate minimizer $\widehat{f} \in \mathcal{F}$ of $\widehat{\rho}^U$ satisfies

$$\begin{aligned}\rho^U(\widehat{f}) &\le \inf_f \widehat{\rho}^U(f) + \varepsilon_{\mathrm{opt}} + r_n \\ &\le \inf_f \rho^U(f) + \varepsilon_{\mathrm{opt}} + 2r_n.\end{aligned}$$

The last inequality takes infima of the pointwise endpoint bound, so no exact minimizer is needed. Uniformity permits all predictors in these claims to be chosen using the same data. □

The selection benchmark is the smallest compatible upper endpoint in the class. For every compatible $P^\star$, the same argument gives $\rho_{\alpha,P^\star}(\widehat{f}) \le \inf_f \rho^U(f) + 2r_n + \varepsilon_{\mathrm{opt}}$. Clipped endpoint errors are at most $2r_n$ on the same event. Convergence in probability follows if $r_n \to 0$ in probability and $\delta_n + \delta_{\nu,n} \to 0$; a fixed failure probability remains fixed.

B.5 Oracle calibration for finite-cell experiments

For the oracle coverage experiment, suppose $p = p_1$ is known and define

$$\begin{aligned} h^\sigma_{f,\eta}(O) &= Sa^{Y^{\mathrm{obs}}}_{f,\eta}(X) + (1-S)b^\sigma_{f,\eta}(X;p(X)), \\ \widehat{\rho}^\sigma_{\mathrm{or}}(f) &= \inf_{\eta\in[0,1]}\{\eta + \beta^{-1}P_n h^\sigma_{f,\eta}\}. \end{aligned} \tag{B.23}$$

This score lies in $[0,1]$ and has mean $m^\sigma_{f,\eta}$; estimating π is unnecessary for this oracle construction.

Proposition B.6 (Finite-cell oracle certificate)**.** *Fix a predictor f independently of the evaluation observations. Suppose P_X is supported on J covariate values $x_1,\dots,x_J$. Let*

$$T(f) = \{0,1\} \cup \{\ell(f(x_j),y) : 1 \le j \le J,\ y \in \{0,1\}\}, \qquad M = |T(f)| \le 2J+2.$$

With probability at least $1-\delta$,

$$\max_{\sigma\in\{L,U\}} |\widehat{\rho}^\sigma_{\mathrm{or}}(f) - \rho^\sigma(f)| \le r_{\mathrm{or}}(\delta) := \frac{1}{\beta}\sqrt{\frac{\log(4M/\delta)}{2n}}. \tag{B.24}$$

Thus the clipped interval obtained from (8) using these oracle estimators and r_{or} contains the entire sharp population interval with probability at least $1-\delta$.

Proof. By the piecewise-affine argument of Proposition B.5, both $h^\sigma_{f,\eta}(O)$ and its expectation have knots only in $T(f)$. The absolute empirical-minus-population difference is maximal at a knot:

$$\sup_{\eta\in[0,1]} |(P_n - P)h^\sigma_{f,\eta}| = \max_{\eta\in T(f)} |(P_n - P)h^\sigma_{f,\eta}|.$$

Hoeffding's inequality and a union bound over the $2M$ endpoint–threshold pairs bound the failure probability at tolerance u by $4Me^{-2nu^2}$. Taking $u = \sqrt{\log(4M/\delta)/(2n)}$ and applying (A.13) proves (B.24); The containment argument in Appendix B.4 gives whole-interval coverage. □

The population-support set $T(f)$ includes losses at cells absent from a particular sample; observed knots $T_n(f)$ suffice for empirical minimization but cannot replace $T(f)$ in this concentration argument. Conditioning covers a predictor trained on independent data. Selection using evaluation data requires a uniform bound: for a fixed finite family $\mathcal{F}_0$, replace M by $\sum_{f\in\mathcal{F}_0}|T(f)|$. Known p supplies the oracle experiment's premise; coverage for learned-nuisance estimators instead requires Theorem 2's controls.

B.6 Revealed-tail learning lower bound

We now examine the difficulty of learning a predictor in the $\Gamma = 1$ submodel, where population risks are identified. The lower-bound construction uses equal tail-cell masses and constant observation probability.

Theorem B.7 (Revealed-tail observation complexity)**.** *Fix $\alpha \in (0,1)$, $\beta = 1-\alpha$, $\pi_0 \in (0,1]$, and $n \ge 1$. For every integer $1 \le d \le n\pi_0\beta$, there exist a binary classification class $\mathcal{F}$ of VC dimension d and a family $\mathcal{P}$ of full-data laws with $\Gamma = 1$ and $\pi(X) = \pi_0$ such that*

$$\begin{aligned} &\inf_{\widehat{f}} \sup_{P^\star\in\mathcal{P}} \mathbb{E}_{(P^\star_{\mathrm{obs}})^{\otimes n}}\left[\rho_{\alpha,P^\star}(\widehat{f}) - \inf_{f\in\mathcal{F}}\rho_{\alpha,P^\star}(f)\right] \\ &\qquad \ge \frac{1}{16}\sqrt{\frac{d}{n\pi_0\beta}}. \end{aligned} \tag{B.25}$$

The infimum ranges over measurable learners taking values in $\mathcal{F}$, including randomized learners.

Proof. We use a direct binary-testing form of the hypercube argument [34]. Fix $\pi_0 \in (0,1]$, $n \geq 1$, and an integer $1 \leq d \leq n\pi_0\beta$. Put

$$\gamma = \frac{1}{8}\sqrt{\frac{d}{n\pi_0\beta}} \leq \frac{1}{8}. \tag{B.26}$$

Let $\mathcal{X} = \{\star, 1, \ldots, d\}$ with $P_X(\star) = 1-\beta$ and $P_X(j) = \beta/d$. For $v \in \{-1,1\}^d$, define $f_v(\star) = 0$ and $f_v(j) = (1+v_j)/2$, and set $\mathcal{F} = \{f_v : v \in \{-1,1\}^d\}$. The class shatters $\{1,\ldots,d\}$, and cannot shatter any larger set because its value at $\star$ is fixed. Its VC dimension is exactly d. Use ordinary zero-one loss $\ell(f(x), y) = \mathbf{1}\{f(x) \neq y\}$.

For each $\theta \in \{-1,1\}^d$, construct a full-data law P_θ with $Y = 0$ at $\star$ and

$$P_\theta(Y = 1 \mid X = j) = \frac{1}{2} + \gamma\theta_j, \qquad j = 1, \ldots, d.$$

Let $S \sim \text{Bernoulli}(\pi_0)$ independently of (X, Y), and observe (X, S, SY). These laws satisfy overlap with $\pi(X) = \pi_0$ and conditional ignorability, hence belong to the $\Gamma = 1$ model. All losses vanish at $\star$. The conditional error within $\{1,\ldots,d\}$ is

$$R_\theta(f_v) = \frac{1}{2} - \gamma + \frac{2\gamma}{d}\sum_{j=1}^{d} \mathbf{1}\{v_j \neq \theta_j\}. \tag{B.27}$$

The full-population loss is Bernoulli with parameter $q_{\theta,v} = \beta R_\theta(f_v) \leq \beta$; the Bernoulli identity (A.5) gives

$$\begin{aligned} \rho_{\alpha,P_\theta}(f_v) &= R_\theta(f_v), \\ \rho_{\alpha,P_\theta}(f_v) - \inf_{f\in\mathcal{F}} \rho_{\alpha,P_\theta}(f) &= \frac{2\gamma}{d}\sum_{j=1}^{d} \mathbf{1}\{v_j \neq \theta_j\}. \end{aligned} \tag{B.28}$$

The minimizing predictor is f_θ.

Let $Q_\theta = P_{\theta,\text{obs}}^{\otimes n}$ and let $\theta^{(j)}$ be obtained by reversing the jth coordinate. For one observation, the two distributions differ only on $\{X = j, S = 1\}$, which has probability $\beta\pi_0/d$ under both laws. Conditioning on (X, S) and using the chain rule for relative entropy therefore gives

$$\begin{aligned} D_{\text{KL}}(Q_\theta \| Q_{\theta^{(j)}}) &= \frac{n\beta\pi_0}{d} D_{\text{KL}}(\text{Bernoulli}(1/2+\gamma) \,\|\, \text{Bernoulli}(1/2-\gamma)) \\ &\leq \frac{16 n\beta\pi_0\gamma^2}{d} = \frac{1}{4}. \end{aligned} \tag{B.29}$$

Here the symmetric Bernoulli divergences both equal $2\gamma\log\{(1+2\gamma)/(1-2\gamma)\} \leq 16\gamma^2$ for $\gamma \leq 1/4$, and independence supplies the factor n. Pinsker's inequality gives

$$\text{TV}(Q_\theta, Q_{\theta^{(j)}}) \leq \sqrt{D_{\text{KL}}(Q_\theta \| Q_{\theta^{(j)}})/2} \leq \frac{1}{2\sqrt{2}} \leq \frac{1}{2}. \tag{B.30}$$

Consider any measurable learner taking values in $\mathcal{F}$, written $\widehat{f} = f_{\widehat{v}}$. For a fixed coordinate j, pair each θ with $\theta_j = 1$ with $\theta^{(j)}$, and write $A_j = \{\widehat{v}_j = 1\}$. The sum of the two testing errors satisfies

$$\begin{aligned} Q_\theta(\widehat{v}_j \neq \theta_j) + Q_{\theta^{(j)}}(\widehat{v}_j \neq \theta_j^{(j)}) &= Q_\theta(A_j^c) + Q_{\theta^{(j)}}(A_j) \\ &\geq 1 - \text{TV}(Q_\theta, Q_{\theta^{(j)}}) \geq \frac{1}{2}. \end{aligned}$$

There are 2^{d-1} such pairs. Averaging over all 2^d vertices gives $2^{-d}\sum_\theta Q_\theta(\widehat{v}_j \neq \theta_j) \geq 1/4$. Sum this inequality over j and use (B.28) to obtain

$$\frac{1}{2^d}\sum_\theta \mathbb{E}_{Q_\theta}\left[\rho_{\alpha,P_\theta}(\widehat{f}) - \inf_{f\in\mathcal{F}} \rho_{\alpha,P_\theta}(f)\right] \geq \frac{2\gamma}{d}\frac{d}{4} = \frac{1}{16}\sqrt{\frac{d}{n\pi_0\beta}}.$$

The supremum over θ is at least this average, for every learner. Taking the infimum over learners proves Theorem B.7. Randomized learners satisfy the same bound: append their independent random seed to each experiment, which leaves the neighboring total variation unchanged, and apply the identical testing argument. □

Only the expected $n\pi_0\beta$ revealed observations in the tail stratum distinguish the vertices. This is an expected excess-CVaR lower bound for $\mathcal{F}$-valued learners, already with fully identified risks at $\Gamma = 1$.

C Revelation Theory and Algorithms

Fix f, $\alpha \in (0,1)$, $\beta = 1-\alpha$, and finite $\Gamma \geq 1$. Use the historical π, p_1, sensitivity bounds $[\underline{p}_\Gamma, \overline{p}_\Gamma]$, and $s_\Gamma = \overline{p}_\Gamma - \underline{p}_\Gamma$. Write $\mu = P_X$ and $\mathbb{E} = \mathbb{E}_\mu$. All kernels and policies are measurable; conditional statements apply on components of positive population mass.

C.1 Retained-information model and compatible laws

Additional revelation $A \in \{0,1\}$ satisfies $A \leq 1-S$, with intensity $\lambda(x) = P(A=1 \mid X=x, S=0) \in [0,1]$. The audited population law has known $p_A(x) = P(Y=1 \mid X=x, A=1)$ satisfying

$$\underline{p}_\Gamma \leq p_A \leq \overline{p}_\Gamma \quad \text{on } \{(1-\pi)\lambda > 0\}. \tag{C.1}$$

Extend p_A within these bounds on its zero-mass region; this has no effect on risks.

Definition C.1 (Retained-sensitivity compatibility)**.** The class $\mathcal{P}^{\mathrm{ret}}_{\Gamma,\lambda}$ consists of laws of (X,S,A,Y) with X-marginal μ and conditional components

(S,A)	$P(S,A \mid X=x)$	$P(Y=1 \mid X=x,S,A)$
$(1,0)$	$\pi(x)$	$p_1(x)$
$(0,1)$	$(1-\pi(x))\lambda(x)$	$p_A(x)$
$(0,0)$	$(1-\pi(x))(1-\lambda(x))$	$q(x)$.

The residual kernel ranges freely over $q \in [\underline{p}_\Gamma, \overline{p}_\Gamma]$; $(S,A) = (1,1)$ has zero mass.

Lemma C.2 (Historical compatibility)**.** *Every admissible q defines a retained law whose marginal belongs to $\mathcal{P}_\Gamma(P_{\mathrm{obs}})$, with original hidden kernel*

$$p_0 = \lambda p_A + (1-\lambda)q \quad \textit{on } \{\pi < 1\}. \tag{C.2}$$

The retained class is nonempty.

Proof. Apply the kernel construction of Appendix A.1 to the table. Its masses sum to one and its selected component is unchanged. Conditional total probability gives (C.2); convexity of the sensitivity interval proves compatibility. Taking $q = \underline{p}_\Gamma$ proves nonemptiness. □

At $\lambda = 0$ the historical class is recovered; at $\lambda = 1$ no residual kernel remains. Membership requires the audited and residual bounds separately: a bound on their mixture p_0 alone does not imply either component bound.

Known randomization $A \perp Y \mid (X, S=0)$ supplies extra information: $p_A = p_0$ on the audited support, and $q = p_0$ wherever residual mass remains. Thus the population audited law identifies p_0 wherever originally hidden cases have positive audit probability. The retained class instead keeps its residual kernel free within the displayed bounds. Finite audits additionally require estimation and uncertainty control for the audited contribution; a positive population audit probability alone supplies no finite-sample precision guarantee.

C.2 Post-revelation endpoints and tail identification value

Use $a^y_{f,\eta}, d_{f,\eta}, b^\sigma_{f,\eta}$ from Section 3, write $s_\Gamma = \overline{p}_\Gamma - \underline{p}_\Gamma$, and set $w_\lambda = (1-\pi)(1-\lambda)$. The known contribution and a residual kernel's threshold objective are

$$K_{f,\eta}(\lambda) = \mathbb{E}[\pi(a^0_{f,\eta} + p_1 d_{f,\eta}) + (1-\pi)\lambda(a^0_{f,\eta} + p_A d_{f,\eta})], \tag{C.3}$$

$$G_q(\eta) = \eta + \beta^{-1}\{K_{f,\eta}(\lambda) + \mathbb{E}[w_\lambda(a^0_{f,\eta} + q d_{f,\eta})]\}. \tag{C.4}$$

Conditioning on the table and optimizing the affine residual term give

$$\begin{aligned} L_{f,\eta}(\lambda) &= \eta + \beta^{-1}\{K_{f,\eta}(\lambda) + \mathbb{E}[w_\lambda b^L_{f,\eta}]\}, \\ U_{f,\eta}(\lambda) &= \eta + \beta^{-1}\{K_{f,\eta}(\lambda) + \mathbb{E}[w_\lambda b^U_{f,\eta}]\}. \end{aligned} \tag{C.5}$$

Proposition C.3 (Sharp retained-sensitivity interval)**.** *The attainable population risks are exactly*

$$\{\rho_{\alpha,Q}(f) : Q \in \mathcal{P}^{\mathrm{ret}}_{\Gamma,\lambda}\} = [\min_{\eta\in[0,1]} L_{f,\eta}(\lambda), \min_{\eta\in[0,1]} U_{f,\eta}(\lambda)].$$

Both threshold minima and endpoint laws are attained.

Proof. On $H = \{x : \pi(x) < 1\}$, the common kernels q^L_f, q^U_f of Appendix A.3 satisfy $a^0_{f,\eta} + q^\sigma_f d_{f,\eta} = b^\sigma_{f,\eta}$ for every $\eta \in [0,1]$. On H^c, the residual weight $w_\lambda = (1-\pi)(1-\lambda)$ is zero. Thus $L_{f,\eta}(\lambda) \le G_q(\eta) \le U_{f,\eta}(\lambda)$ for every admissible q, with equality at the corresponding common kernel. Taking minima proves the endpoint claims; continuity of each hinge objective on $[0,1]$ gives attainment. For $q_t = (1-t)q^L_f + tq^U_f$,

$$\sup_\eta |G_{q_t}(\eta) - G_{q_s}(\eta)| \le |t-s|\beta^{-1}\mathbb{E}[w_\lambda s_\Gamma].$$

The infimum inequality (A.13) makes the optimized risk continuous in t. This admissible path joins the endpoints and fills their interval. □

Proposition C.4 (Exact revelation value)**.** *With*

$$\tau_{f,\eta} = (1-\pi)s_\Gamma|d_{f,\eta}|/\beta, \tag{C.6}$$

$$W_{f,\eta}(\lambda) := U_{f,\eta}(\lambda) - L_{f,\eta}(\lambda) = \mathbb{E}[(1-\lambda)\tau_{f,\eta}] = W_{f,\eta}(0) - \mathbb{E}[\lambda\tau_{f,\eta}], \tag{C.7}$$

every bounded measurable h and feasible t satisfy

$$W_{f,\eta}(\lambda + th) - W_{f,\eta}(\lambda) = -t\mathbb{E}[h\tau_{f,\eta}]. \tag{C.8}$$

Proof. Subtract (C.5): K cancels and $b^U - b^L = s_\Gamma|d|$. Substitution gives the perturbation identity, including one-sided feasible directions. Since $0 \le \tau \le \beta^{-1}$, all terms are integrable. The fixed-threshold reduction is independent of p_A, even when that law varies with the policy. □

C.3 Minimum-cost precision and the exact frontier

Fix f, η, Γ, abbreviate $\tau = \tau_{f,\eta}$, and let $c \ge 0$ be measurable with $\mathbb{E}[c] < \infty$. Set $T = \mathbb{E}[\tau] = W_{f,\eta}(0)$, $a = (T-\varepsilon)_+$ for $\varepsilon \ge 0$, and $F = \mathbb{E}[\tau \mathbf{1}_{\{c=0\}}]$. By (C.7), the precision constraint is $\mathbb{E}[\tau\lambda] \ge a$.

Proof of Theorem 3. We establish attainment and the primal–dual identity

$$\mathrm{Cert}_{f,\eta}(\varepsilon) = \min_{0\le\lambda\le1}\{\mathbb{E}[c\lambda] : \mathbb{E}[\tau\lambda] \ge a\} = \sup_{u\ge0}\{ua - \mathbb{E}[(u\tau - c)_+]\}. \tag{C.9}$$

Pointwise, $(c - u\tau)\lambda \ge -(u\tau - c)_+$, so every feasible policy obeys

$$\mathbb{E}[c\lambda] \ge u\mathbb{E}[\tau\lambda] - \mathbb{E}[(u\tau - c)_+] \ge ua - \mathbb{E}[(u\tau - c)_+]. \tag{C.10}$$

The following four cases attain this lower bound, or its limit.

No reduction ($a = 0$). Take $\lambda = 0, u = 0$, giving zero cost; this includes $T = 0$.

Free reduction ($0 < a \le F$). Take $\lambda = (a/F)\mathbf{1}_{\{c=0,\tau>0\}}$ and $u = 0$. The reduction is exactly a at zero cost, including when the free component has atoms.

Interior reduction ($F < a < T$). On $D = \{c > 0, \tau > 0\}$, use the finite measure $\nu(B) = \int_B \tau\, d\mu$ and $r = c/\tau$. Put $b = a - F \in (0, \nu(D))$ and choose

$$u_* = \inf\{v \ge 0 : \nu(D \cap \{r \le v\}) \ge b\}. \tag{C.11}$$

Because $r \in (0,\infty)$ ν-almost everywhere, $0 < u_* < \infty$. Continuity of a finite measure along monotone sets gives $H := \nu(D \cap \{r < u_*\}) \le b \le H + J$, where $J := \nu(D \cap \{r = u_*\})$. Set $\theta = (b - H)/J$ if $J > 0$ and $\theta = 0$ otherwise (then $H = b$). The policy

$$\lambda_* = \mathbf{1}_{\{c=0,\tau>0\}} + \mathbf{1}_{D\cap\{r<u_*\}} + \theta\mathbf{1}_{D\cap\{r=u_*\}} \tag{C.12}$$

achieves reduction $F + H + \theta J = a$ and equality in the pointwise dual bound. Its cost equals $u_*a - \mathbb{E}[(u_*\tau - c)_+]$. It ranks by τ/c with threshold $1/u_*$; fractional ties handle atoms without an atomlessness assumption.

Full reduction ($a = T > F$). Feasibility forces $\mathbb{E}[\tau(1-\lambda)] = 0$, hence $\lambda = 1$ almost surely on $\{\tau > 0\}$. The minimum cost is $\mathbb{E}[c\mathbf{1}_{\{\tau>0\}}]$, attained by revealing exactly that set, while

$$uT - \mathbb{E}[(u\tau - c)_+] = \mathbb{E}[\min\{u\tau, c\}] \uparrow \mathbb{E}[c\mathbf{1}_{\{\tau>0\}}]$$

by monotone convergence. Thus the dual supremum agrees even without a finite maximizer. Its shortfall is $\mathbb{E}[(c - u\tau)_+\mathbf{1}_{\{\tau>0\}}]$, so a finite optimizer exists exactly when $c \le u\tau$ almost surely on $\{\tau > 0\}$ for some finite u. Full and free reduction correspond to value-density thresholds 0 and ∞. Zero-value locations can always be left unselected; if $F = T$ the whole frontier is free. These cases exhaust $0 \le a \le T$. □

C.4 From threshold precision to optimized CVaR precision

Denote the endpoints in Proposition C.3 by $\underline{\rho}^{\mathrm{ret}}_{\alpha,\Gamma,\lambda}(f)$ and $\overline{\rho}^{\mathrm{ret}}_{\alpha,\Gamma,\lambda}(f)$, and their difference by $I^{\mathrm{ret}}_{\Gamma,\lambda}(f)$.

Proposition C.5 (Controlling the optimized CVaR interval)**.** *For the fixed predictor,*

$$0 \le I^{\mathrm{ret}}_{\Gamma,\lambda}(f) \le \sup_{\eta\in[0,1]} W_{f,\eta}(\lambda) = W_{f,0}(\lambda). \tag{C.13}$$

Theorem 3 at $\eta = 0$ therefore minimizes cost for the sufficient condition $W_{f,0}(\lambda) \le \varepsilon$.

Proof. If η_L minimizes L, then $\min_\eta U - \min_\eta L \le U_{f,\eta_L} - L_{f,\eta_L} = W_{f,\eta_L}$; nonnegativity follows from ordered endpoints. For $v = \max_y \ell(f(x), y)$ and $w = \min_y \ell(f(x), y)$,

$$|d_{f,\eta}(x)| = (v-\eta)_+ - (w-\eta)_+ = (v - \max\{w, \eta\})_+.$$

This decreases in η and equals $v - w$ at zero. Integration against $(1-\lambda)(1-\pi)s_\Gamma/\beta \ge 0$ proves the supremum identity. □

Direct optimized-width design. Fix an admissible set Λ_{adm} of measurable revelation intensities and a specified family of population audit laws $\{P^\lambda : \lambda \in \Lambda_{\mathrm{adm}}\}$ sharing one (X, S, Y)-marginal. Each P^λ has intensity λ and satisfies Definition C.1. Set $p^\lambda_A(x) = P^\lambda(Y = 1 \mid X = x, A = 1)$, with an extension within the sensitivity bounds on zero-mass regions. Write $I^{\mathrm{ret}}_{\Gamma,\lambda}(f; p^\lambda_A)$ for the width obtained using this kernel in (C.3), retaining the original free residual bounds. For the cost density c above and $\varepsilon \ge 0$, define

$$\mathrm{Cost}^{\mathrm{CVaR}}_f(\varepsilon) := \inf_{\lambda\in\Lambda_{\mathrm{adm}}} \left\{\mathbb{E}[c\lambda] : I^{\mathrm{ret}}_{\Gamma,\lambda}(f; p^\lambda_A) \le \varepsilon\right\}, \tag{C.14}$$

with $\inf \varnothing = +\infty$. The audited contribution can affect the separate threshold minima even though it cancels from $W_{f,\eta}(\lambda)$.

Strictness of the sufficient bound. Take constant X, zero-one loss, $f = 0$, $\pi = p_1 = p_A = 1/2$, $\lambda = 1/2$, $\Gamma = 2$, and $\beta = 1/4$. The original bounds are $[1/3, 2/3]$. Under the retained lower and upper endpoint laws, the probabilities r_L and r_U of unit loss $L_f = 1$ are

$$r_L = \tfrac{1}{4} + \tfrac{1}{8} + \tfrac{1}{12} = \tfrac{11}{24}, \qquad r_U = \tfrac{1}{4} + \tfrac{1}{8} + \tfrac{1}{6} = \tfrac{13}{24}.$$

Both exceed β, so (A.5) gives optimized endpoints one and $I^{\mathrm{ret}}_{\Gamma,\lambda}(f) = 0$, although $W_{f,0}(\lambda) = \{(1/2)(1/2)/(1/4)\}(2/3 - 1/3) = 1/3$. Thus the sufficient criterion can require revelation when optimized risk is already identified. Predictor reoptimization and estimated acquisition values require their own statistical or sequential analysis.

C.5 Finite-cell computation and operational cost units

For cells $j = 1, \ldots, K$ with masses $\mu_j > 0$, write $\tau_j = \tau_{f,\eta}(j)$, $c_j = c(j)$, and set $v_j = \mu_j\tau_j$, $w_j = \mu_j c_j$, $T = \sum_j v_j$, $a = (T - \varepsilon)_+$, and $F = \sum_{j:c_j=0} v_j$. The exact fractional algorithm is:

1. If $a = 0$, reveal nothing. If $0 < a \le F$, reveal fraction a/F of the zero-cost, positive-value component; both cases cost zero.
2. Otherwise reveal all useful free cells. Order the positive-cost, positive-value cells by decreasing τ_j/c_j; set $R_k = \sum_{j\le k} v_j$, $C_k = \sum_{j\le k} w_j$, and $R_0 = C_0 = 0$.

3. For $b = a - F$, let $k = \min\{j : R_j \geq b\}$. Reveal the cells before k, fraction $\theta = (b - R_{k-1})/v_k$ of cell k, and none after it. Then

$$\mathrm{Cert}_{f,\eta}(\varepsilon) = C_{k-1} + \theta w_k = C_{k-1} + \frac{b - R_{k-1}}{\tau_k / c_k}. \tag{C.15}$$

Tied ratios may be ordered arbitrarily; full reduction gives $\theta = 1$ in the last useful cell. If there are no positive-cost useful cells, the first step covers every feasible target. This is Theorem 3's policy, so the frontier is piecewise linear. For fixed cell policies and audited laws, the piecewise-affine argument of Proposition B.5 optimizes both retained objectives over $\{0, 1\} \cup \{\ell(f(j), y) : 1 \leq j \leq K,\ y \in \{0, 1\}\}$.

If a hidden label costs $\kappa(x) \geq 0$, the target-population cost density and operational value density are

$$c(x) = (1 - \pi(x))\kappa(x), \qquad \mathbb{E}[c\lambda] = \mathbb{E}[(1 - \pi)\lambda\kappa], \tag{C.16}$$

$$\frac{\tau_{f,\eta}(x)}{c(x)} = \frac{s_\Gamma(x)|d_{f,\eta}(x)|}{\beta\kappa(x)} \quad \text{when } (1 - \pi)\kappa > 0. \tag{C.17}$$

The hidden-mass factor cancels; unit label cost ranks already-hidden cases by $s_\Gamma|d|/\beta$. Cell masses likewise cancel from v_j/w_j. The algorithm allocates divisible population mass. Whole-individual batches add integer constraints and sampling variation. Sequential retraining changes losses and acquisition values; confidence guarantees across adaptive rounds additionally require control of newly revealed contributions, acquisition history, and stopping.

D Experimental Protocols and Results

This appendix specifies the experiments in Section 6: finite-cell identification and estimation, historical applicant assessment, controlled masking, and targeted revelation. All results use saved outputs; Appendix D.7 identifies the accompanying provenance.

D.1 Evaluation targets, interval construction, and aggregation

Write $\beta = 1 - \alpha$, $\rho^L = \underline{\rho}_{\alpha,\Gamma}$, $\rho^U = \overline{\rho}_{\alpha,\Gamma}$, and $I_\Gamma = \rho^U - \rho^L$. Endpoints minimize their respective threshold objectives. Finite-support minima are found exactly at 0, 1, and all label-specific loss breakpoints, including when estimated hinge coefficients are signed. Sampling and lending studies use Brier loss $(f - y)^2$; population sweeps and the synthetic frontier use absolute loss.

Study	Target	Metric	Replications/ aggregation
Finite-cell sampling	Sharp population interval; its endpoints	Oracle whole-interval containment; learned-endpoint RMSE	60 worlds × 80 samples per n; means/RMSE
Historical applicants	Applicant-level regression-extension sensitivity interval	Estimated endpoints and strict comparisons	One cohort; five-fold cross-fitting
Controlled masking	Full-outcome CVaR of each fitted predictor	Reference coverage; comparison errors and unresolved decisions	20 masks; counts and mean widths
Fixed revelation	Fixed-threshold retained gap $W_{f,\eta}$	Remaining gap; cost to target	120 synthetic worlds or 20 masks; within-run normalization, then medians/IQR
Sequential audits	Estimated optimized retained width; held-out predictive loss	Width, mean Brier loss, CVaR; paired TIV–random differences	20 masks; medians and descriptive IQR

For fixed f, η, the revelation target is $W_{f,\eta}(\lambda) = \mathbb{E}[(1 - \lambda)\tau_{f,\eta}]$, where $\tau_{f,\eta} = (1 - \pi)s_\Gamma|d_{f,\eta}|/\beta$. It differs from the optimized width, whose endpoints are minimized separately (Appendix C.4). Oracle containment, masking-reference coverage, and sequential plug-in precision have different

calibration premises. Fixed-audit reductions normalize within each run before aggregation; sequential widths aggregate the recorded widths. Ratios of separately aggregated values need not equal aggregated relative reductions.

D.2 Finite-cell identification and learned-endpoint accuracy

Generator and estimators. Each of 60 simulated populations has covariate support $\mathcal{X}_0$ consisting of 30 cells, with Dirichlet$(1, \ldots, 1)$ masses, independent $\pi \sim U[.2, .95]$ and $p_1 \sim U[.05, .95]$, and $p_0 \mid p_1 \sim U[q_-(p_1), q_+(p_1)]$ at $\Gamma = 2$.

Fix $f = \text{clip}_{[0,1]}(p_1 + N(0, .15^2))$ within each world, use Brier loss and $\alpha = .75$, and compute exact endpoints and generating-law CVaR on $T_f = \{0, 1\} \cup \{f(x)^2, (1 - f(x))^2 : x \in \mathcal{X}_0\}$. For each $n \in \{1000, 3000, 10000, 30000, 100000\}$, draw 80 samples per world (4,800 replications). The oracle uses known p_1 and the bounded score of Appendix B.5; its endpoints are expanded by

$$r_{\text{or}} = \frac{1}{\beta}\sqrt{\frac{\log(4|T_f|/\delta)}{2n}}, \qquad \delta = .05,$$

then clipped to $[0, 1]$. Proposition B.6 gives whole-interval containment with probability at least .95 for this fixed predictor. Saved generating-law CVaR coverage also equals one at every reported sample size.

On the same records, randomly assign each observation to one of two folds. In a given training cell, let N_{11} count records with $(S, Y) = (1, 1)$, let N_{10} count records with $(S, Y) = (1, 0)$, and let N be the total number of training records in that cell. Estimate

$$\widehat{p}_1 = \frac{N_{11} + 1/2}{N_{11} + N_{10} + 1}, \qquad \widehat{\pi} = \frac{N_{11} + N_{10}}{N},$$

clipped to $[.01, .99]$ and $[.1, 1]$, respectively; an empty training cell receives propensity $1/2$. Evaluate the locally robust score using the other fold's fits. The resulting RMSEs measure endpoint accuracy. Learned-nuisance coverage requires a separate calibration argument.

n	Sharp width	Radius	Outer width	Coverage	RMSE upper	RMSE lower
1,000	.0949	.260	.614	1.000	.0361	.0425
3,000	.0949	.150	.395	1.000	.0171	.0160
10,000	.0949	.082	.259	1.000	.0087	.0076
30,000	.0949	.047	.190	1.000	.0049	.0043
100,000	.0949	.026	.147	1.000	.0027	.0023

Table 1: Repeated-sampling decomposition over 4,800 replications per sample size. Widths and oracle radii are means; coverage is the proportion of oracle outer intervals containing both exact endpoints. RMSE uses two-fold learned-nuisance estimates on those same sampled records.

Observed-tail scaling. For the $d = 16$ tail-cell construction of Appendix B.6, use 240 replications at each of 54 settings: $n \in \{600, 900, 1400, 2200, 3500, 5500\}$, $\beta \in \{.10, .15, .25\}$, and $\pi_0 \in \{.20, .35, .55\}$. Set $\gamma = \min\{.20, .16\sqrt{d/(n\beta\pi_0)}\}$ and recover each cell's sign by majority vote among revealed tail labels, assigning ties the positive sign. The metric is 2γ times the incorrectly recovered fraction, equal to excess CVaR for this construction. Its least-squares guide against $1/\sqrt{n\pi_0\beta}$ has slope .5030, intercept $-.0018$, and $R^2 \simeq .998$; this illustration does not establish a general matching upper bound for Appendix B.6.

Population sweeps. The five-region absolute-loss design has region masses $\mu_j = P(X = j)$ and full-data outcome probabilities $p_{\text{full},j} = P(Y = 1 \mid X = j)$, for $j = 1, \ldots, 5$. It fixes

$$\mu = (.45, .15, .15, .10, .15), \quad p_{\text{full}} = (.04, .75, .10, .40, .50),$$
$$s_1 = (.80, .03, .80, .80, .80), \quad s_0 = (.80, .60, .80, .80, .80),$$
$$f = (.05, .25, .10, .40, .50), \quad s_y = P(S = 1 \mid X, Y = y).$$

Exact finite-support optimization evaluates 60 tail levels on $[.55, .95]$ at $\Gamma = 50$, and 60 common historical observation probabilities on $[.05, 1]$ at $\alpha = .85, \Gamma = 50$, holding the original p_1 fixed in the latter sweep. An archived sweep uses 60 log-spaced $\Gamma \in [1, 100]$ at $\alpha = .85$; values below the generating sensitivity shift do not claim generating-law coverage.

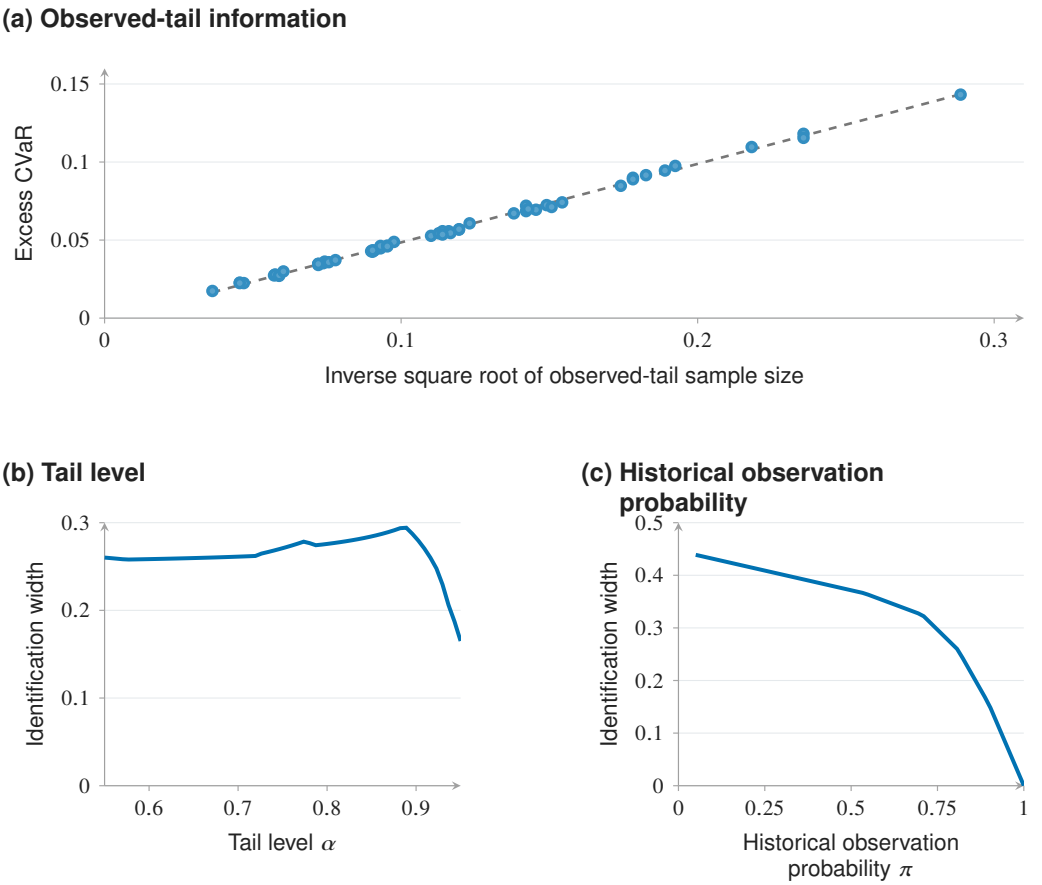


Figure 5: **Observation scarcity and sensitivity shape tail-risk precision.** (a) Recorded simulations of the Appendix B.6 learning construction, with a least-squares guide against $(n\pi_0\beta)^{-1/2}$. Here $n\pi_0\beta$ is the expected observed-tail sample count. (b–c) Exact identification widths for the fixed five-region design as the tail level or historical observation probability varies. Width can vary nonmonotonically with α. Lines join recorded evaluations.

D.3 Historical LendingClub applicant assessment

Cohort and shared preprocessing. From the public 2007–2018 accepted/rejected files [13], retain loans issued and applications rejected in 2012. Mature issued outcomes are charged-off/default ($Y = 1$) or fully paid ($Y = 0$), including corresponding former-credit-policy statuses; exclude nonmature outcomes. The saved cohort has 390,644 applications: 53,367 issued loans (8,644 defaults) and 337,277 rejections. Repayment under issuance is unobserved for rejected applications.

Features are recorded amount, debt-to-income ratio (DTI), FICO credit-score interval midpoint or recorded risk score, employment years, state, three-digit ZIP prefix, and normalized purpose; exclude accepted-only pricing/repayment fields, policy code, and calendar month. Within each training sample, median-impute numeric fields with missingness indicators, most-frequent-impute categorical fields, and one-hot encode categories with minimum frequency 25. Standardize numeric fields for logistic regression. Preprocessing, logistic regression, and stratified splitting use scikit-learn [26].

Eligibility, measurement, and extrapolation. The 2012 FICO minimum was 660 [19]. Source checks identify 180,903 below-660 applications (46.31%) and 23,097 missing standardized scores, with no issued loans in either group. All 53,367 issued loans lie among the remaining 186,644 records at or above 660. This structural restriction does not make FICO eligibility sufficient for joint covariate support or other underwriting requirements.

Recorded fields also differ [18]: rejected applications record FICO, whereas issued loans use an origination-score interval midpoint, without common coarsening; issued `loan_amnt` can reflect credit-department reductions, whereas rejected `Amount Requested` is the requested total. Issued purpose categories and rejected free-text titles are regex-normalized; dates refer to funding month and application date, respectively. Thus these are not identically measured predecision covariates.

In training fold $-k$, fit the issued-outcome regression $\widehat{r}_{-k}$ using the boosting settings below, evaluate every held-out record, and specify

$$r_i = \mathrm{clip}_{[.01,.99]}\, \widehat{r}_{-k(i)}(X_i), \qquad q_i \in [q_-(r_i), q_+(r_i)]. \tag{D.1}$$

On unsupported recorded covariates, r_i is an extrapolated reference, not an identified selected-label probability. Objectives use (5) with $p = r_i$, clipped fitted propensity, and the selected-observation residual correction. These residual-adjusted model-based estimates do not satisfy the population-overlap premise of Theorem 2 on the full cohort.

Models, cross-fitting, and optimization. Candidates use standard classification objectives: ℓ_2 logistic regression ($C = 1$, L-BFGS, 1000-iteration limit), default-weighted logistic regression (default class weight 2), and XGBoost [6] (450 trees, depth 5, learning rate .04, minimum child weight 10, row/column subsampling .85, ℓ_2 penalty 2, histogram trees, recorded CUDA configuration). Both nuisance classifiers use the boosting settings with separately fitted preprocessing and distinct seeds: p_1 uses revealed outcomes and π predicts S from all training covariates. Shuffled five-fold cross-fitting is stratified by S; clip $\widehat{p}_1$ to $[.01, .99]$ and $\widehat{\pi}$ to $[.02, .98]$. Evaluate Brier loss at $\alpha \in \{.90, .95\}$ and $\Gamma \in \{1, 1.25, 1.5, 2, 3, 5, 8, 12\}$.

The corrected objective can be nonconvex because hinge coefficients can be signed. Enumerate 0, 1, and both label-specific losses, $f(X_i)^2$ and $(1 - f(X_i))^2$, at every observed covariate X_i, including records whose actual outcomes are hidden. Deduplicate and sort increasingly; among exactly equal computed minima, choose the smallest threshold, without numerical tie tolerance. Clip endpoints to $[0, 1]$; if they cross, replace both by their midpoint. These clipping and crossing conventions are computational operations and do not supply statistical coverage.

Fold enlargement and diagnostics. At each endpoint's selected threshold, compute five fold-specific objectives and divide their sample standard deviation by $\sqrt{5}$ to obtain an empirical standard error. Let $t_{.975,4}$ be the .975 quantile of a Student-t distribution with four degrees of freedom. Subtract this multiplier times the lower-endpoint standard error from the raw lower endpoint and add the corresponding upper-endpoint quantity to the raw upper endpoint, before clipping or crossing repair. Then clip both limits to $[0, 1]$ and order them by minimum and maximum. This empirical diagnostic uses dependent fold-specific quantities because their training samples overlap; calibrated coverage requires the conditions and radius of Theorem 2.

Historical propensity AUC is .9944 and Brier score .0201; raw propensities are below .02 for 75.26% and above .98 for 1.80% of applicants, and inverse clipped propensities are at most 50, indicating substantial extrapolation. Subgroups reuse full-cohort cross-fitted predictions and require at least $\max\{250, \lfloor .002N \rfloor\}$ applications and 50 issued loans. At $\Gamma = 2, \alpha = .95$, boosting width is .080 for DTI above 30% (2.7% issued) and .032 for FICO 700–739 (30.3% issued).

α	Predictor	Accepted-only	$\Gamma = 1$	$\Gamma = 2$	$\Gamma = 3$
.95	Gradient boosting	.795	.733	[.703,.759]	[.682,.773]
.95	Logistic	.801	.805	[.770,.831]	[.745,.843]
.95	Default-weighted	.661	.813	[.782,.833]	[.757,.841]
.95	Constant $1/2$	.250	.250	[.250,.250]	[.250,.250]
.90	Gradient boosting	.740	.679	[.625,.715]	[.575,.732]
.90	Logistic	.741	.721	[.669,.760]	[.631,.778]
.90	Default-weighted	.579	.720	[.679,.747]	[.646,.759]
.90	Constant $1/2$	.250	.250	[.250,.250]	[.250,.250]

Table 2: Historical applicants: accepted-only CVaR and model-based applicant-level sensitivity intervals. Values other than the exact constant reference are estimates. At $\Gamma = 1$ the reference model fixes the hidden probabilities. Default-weighted denotes logistic regression with default class weight 2.

For $f \equiv 1/2$, both label-specific Brier losses equal $1/4$. Mean loss and every CVaR endpoint therefore equal $1/4$, and $d_{f,\eta} = \mathrm{TIV} = 0$ for every threshold. This row provides an exact predictor reference; the acquisition policies compare audits for a fixed or sequentially refitted boosting predictor.

Off-grid breakdown. For an accepted-only preferred f and comparator g, set $G(\Gamma) = \widehat{\rho}^{L}_{\Gamma}(g) - \widehat{\rho}^{U}_{\Gamma}(f)$. At the first grid value Γ_+ with $G(\Gamma_+) \le 0$, if its predecessor Γ_- has positive gap, interpolate:

$$\log \widehat{\Gamma}_{\mathrm{br}} = \log \Gamma_- + \frac{G(\Gamma_-)}{G(\Gamma_-) - G(\Gamma_+)} (\log \Gamma_+ - \log \Gamma_-).$$

These values interpolate the recorded grid. Boosting versus logistic gives approximate breakdown 2.24 at $\alpha = .95$, with first failed grid value 3, and 1.40 at $\alpha = .90$, with first failed value 1.5. Default-weighted versus logistic gives approximately 1.01 at $\alpha = .90$, with first failed value 1.25. The other three accepted-only orderings fail already at $\Gamma = 1$. A comparison with no crossing through 12 is recorded as right-censored above the grid.

D.4 Controlled masking: empirical coverage and model comparisons

To evaluate coverage against known full-outcome risks, we use the 53,367 mature issued loans as the target population and independently draw 20 masks from

$$\begin{aligned} \operatorname{logit} P(S = 1 \mid X, Y) &= c + g(X) - Y \log 3, \\ g(X) &= \operatorname{clip}_{[-3,3]}\big\{.65z(\text{score}) - .35z(\text{DTI}) \\ &\qquad + .15z(\log(1 + \text{amount}))\big\}. \end{aligned}$$

Each z standardizes the median-imputed pseudo-population feature. Solve for expected revealed share .45 ($c = -.0423744$). Bayes' rule gives $\operatorname{odds}(p_0)/\operatorname{odds}(p_1) = 3$, hence $\Gamma_0 = 3$; boundary outcome probabilities coincide across hidden/revealed groups as in Appendix A.1. This validates artificial masking of mature issued loans, not outcomes for historically rejected applicants.

All three predictors share shuffled five-fold splits stratified by imposed S; fitting, clipping, threshold search, and fold enlargement follow Appendix D.3. Reference CVaR uses every known outcome with that predictor's cross-fitted predictions. Table 3 counts intervals containing this full-outcome reference risk. Table 4 checks strict separation of plug-in intervals against reference risk orderings. Three unordered predictor pairs per mask and 20 masks give 60 comparisons at each tail level, or 120 across both levels.

Figure 7 retains the complete assumed-Γ grid; plug-in coverage is zero at every tested $\Gamma \le 2$. Main-figure pooling averages 20 mask-level means over three predictors, giving 60 equally weighted predictor–mask rows from 20 independent masks. Fold enlargement is an empirical calibration diagnostic; the class-wide calibrated statement is Theorem 2.

α	Predictor	Plug-in	Enlarged	Mean width	Enlarged width
.90	Gradient boosting	11/20	19/20	.401	.434
.90	Logistic	13/20	20/20	.401	.434
.90	Default-weighted	14/20	20/20	.332	.365
.95	Gradient boosting	14/20	19/20	.070	.089
.95	Logistic	12/20	20/20	.066	.084
.95	Default-weighted	14/20	20/20	.110	.139

Table 3: Controlled masking at $\Gamma = \Gamma_0 = 3$. The middle columns count intervals covering the full pseudo-population reference CVaR over 20 masks; the last two columns are mean widths. Fold enlargement follows the empirical diagnostic in Appendix D.3.

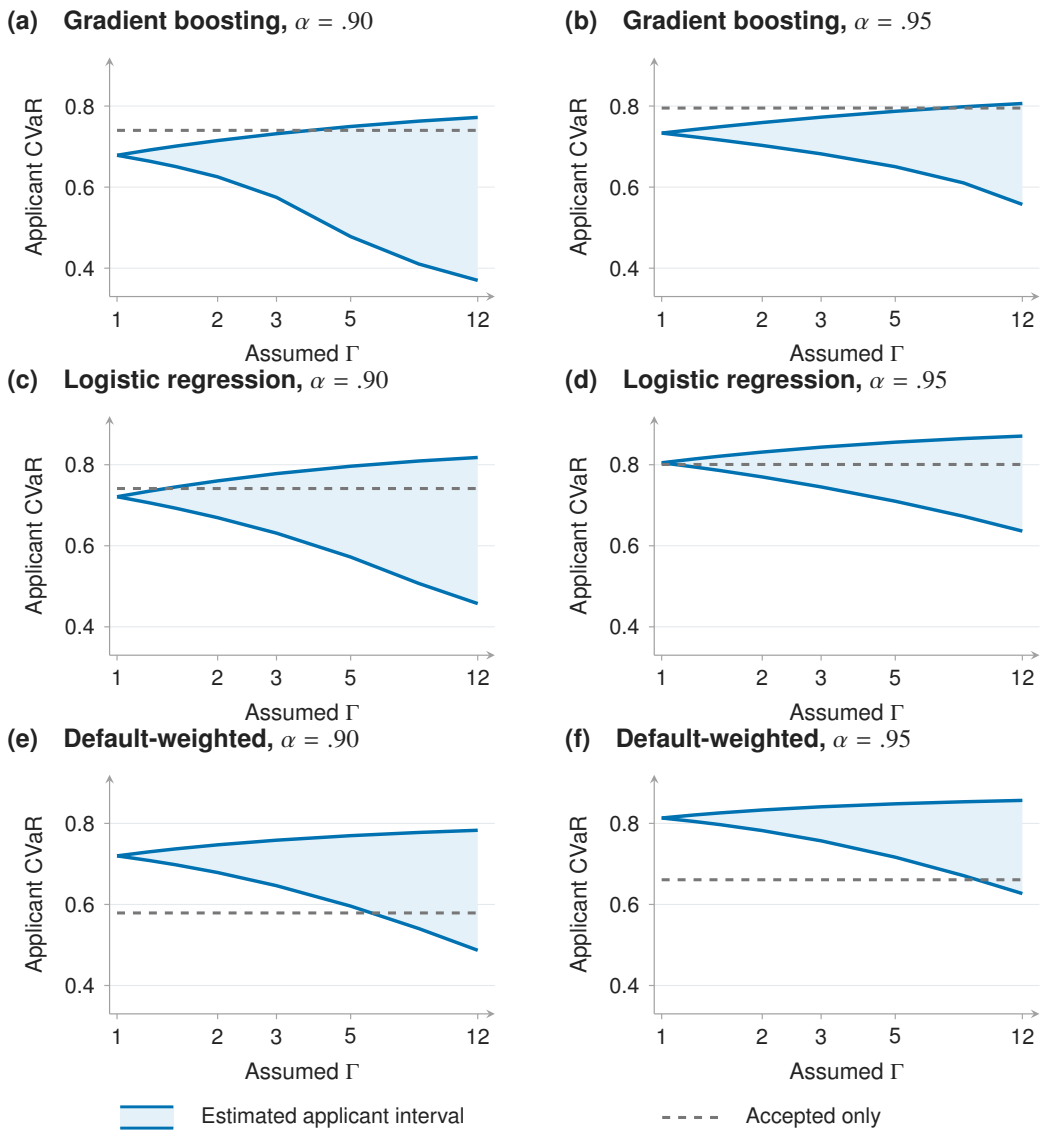


Figure 6: **Complete applicant-level sensitivity analysis.** All three predictors at $\alpha = .90$ and .95 over eight recorded sensitivity levels. Blue boundaries and shading show estimated sensitivity endpoints under the applicant regression extension of Appendix D.3. Gray dashed lines show accepted-only CVaR. Horizontal axes are logarithmic and vertical limits are shared. Default-weighted denotes default-weighted logistic regression.

Γ	α	Comparisons	Resolved	Incorrect	Unresolved
3	.90	60	0	0	60
3	.95	60	37	0	23
1	Both levels	120	120	16	0

Table 4: Model comparisons under controlled masking. Strict separation of plug-in intervals declares an ordering, whose correctness is checked against known-outcome reference CVaRs. Counts give resolved pairs, incorrect declared orderings, and unresolved pairs. Each tail level includes three unordered predictor pairs in each of 20 masks.

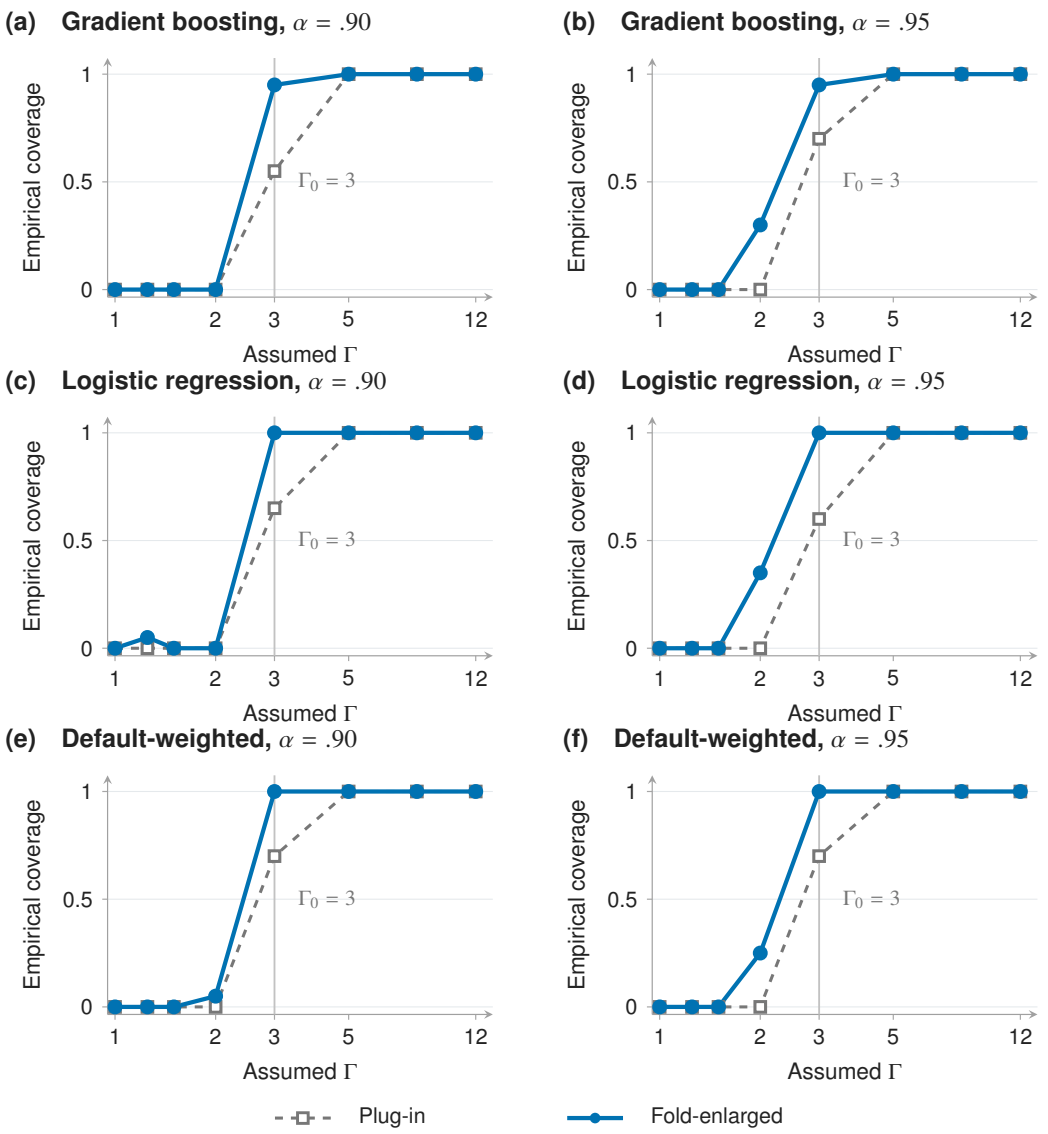


Figure 7: **Empirical coverage by predictor and tail level.** Each proportion uses 20 controlled masks. Blue circles show fold-enlarged intervals and hollow gray squares plug-in intervals; the vertical reference marks the masking mechanism's sensitivity $\Gamma_0 = 3$. Predictors share the same 20 masks, so the predictor-specific proportions are dependent. Default-weighted denotes default-weighted logistic regression.

D.5 Synthetic fixed-threshold cost–precision frontier

We next evaluate the fixed-threshold cost–precision frontier of Theorem 3 in synthetic populations with known outcome probabilities and acquisition costs.

Generator. Use 120 worlds of 120 cells, $\alpha = .85$, $\Gamma = 12$, absolute loss, $f = p_1$, and $\eta = .5$. Masses are Dirichlet$(.75, \dots, .75)$. Independently draw $z_j, e_j, e_j^{\text{out}}, e_j^{\text{in}} \sim N(0,1)$ across cells/noise types. Hidden-tail cells satisfy $z_j + .35e_j$ at or above its empirical .65 quantile (NumPy default linear interpolation). Outside/inside these cells, respectively, set

$$p_{1j} = \text{sigmoid}(-1.35 + .75z_j + .45e_j^{\text{out}}), \qquad p_{1j} = \text{sigmoid}(-2.05 + .35z_j + .30e_j^{\text{in}}),$$

clipped to $[.04, .72]$, and draw $\pi_j \sim U[.62, .95]/U[.02, .16]$. Add to the log-odds of p_{1j} a shift $N(0, .18^2)/U[.8 \log 10, \log 10]$ to obtain p_{0j}, clipped to $[.02, .98]$. Costs are $c_j = U_j + B_j V_j$ with $U_j \sim U[.45, 1.75]$, $B_j \sim \text{Bernoulli}(.20)$, and $V_j \sim U[.75, 1.75]$. Primitive draws are mutually independent and independent of masses; hidden-cell membership induces the specified dependence of π_j and log-odds shifts on the normal scores.

The generated p_{0j} specifies the hidden-outcome distribution in cell j. The fixed-threshold frontier follows from $W_{f,\eta}(\lambda) = \sum_j \mu_j(1-\lambda_j)\tau_{f,\eta}(j)$ and cost $\sum_j \mu_j c_j \lambda_j$. The frontier calculation uses $p_1, \pi, f, \eta, \Gamma, \mu, c$; the generated p_0 does not enter these formulas.

Allocations and equivalent baselines. Here c_j is population cost density: allocations cost $\sum_j \mu_j c_j \lambda_j$, with full cost $\sum_j \mu_j c_j$. Appendix C.5's fractional-knapsack solution uses useful zero-cost cells first, then decreasing τ_j / c_j on positive-cost/value cells (this generator has positive costs). A fractional final cell implements 51 budgets from zero to full cost in .02 increments. TIV/cost and the exact frontier share this ordering and all saved coordinates.

Seven baselines rank by raw independent $U[0,1]$ random score, cell mass, negative revelation probability, selected-label expected absolute loss, $p_1(1-p_1)$, sensitivity width, or absolute tail-excess contrast $|d_{f,\eta}|$. Only TIV/frontier divide their scores by cost. Each world's random scores are independent of generator draws and reused across budgets. Decreasing NumPy default sorting (`argsort(score)[::-1]`) imposes no stable secondary tie rule, including clipping-induced ties. For $v_j = p_{1j}(1-p_{1j})$, predicted loss is $2v_j$, predictive uncertainty is v_j, and

$$s_\Gamma(p_{1j}) = \frac{(\Gamma^2 - 1)v_j}{\Gamma + (\Gamma - 1)^2 v_j}. \tag{D.2}$$

The last expression increases strictly in v_j for $\Gamma > 1$, so these three baselines share a mathematical weak ordering and their saved curves coincide.

Normalize remaining ambiguity by its initial value within each world before taking medians/IQR. Cost-to-target is the first recorded budget reaching 25% of initial ambiguity, divided by TIV's corresponding cost in that world; there is no interpolation between budgets. Normalization requires positive initial ambiguity, full cost, and TIV target cost. This measures divisible population revelation at fixed f, η, Γ (Theorem 3).

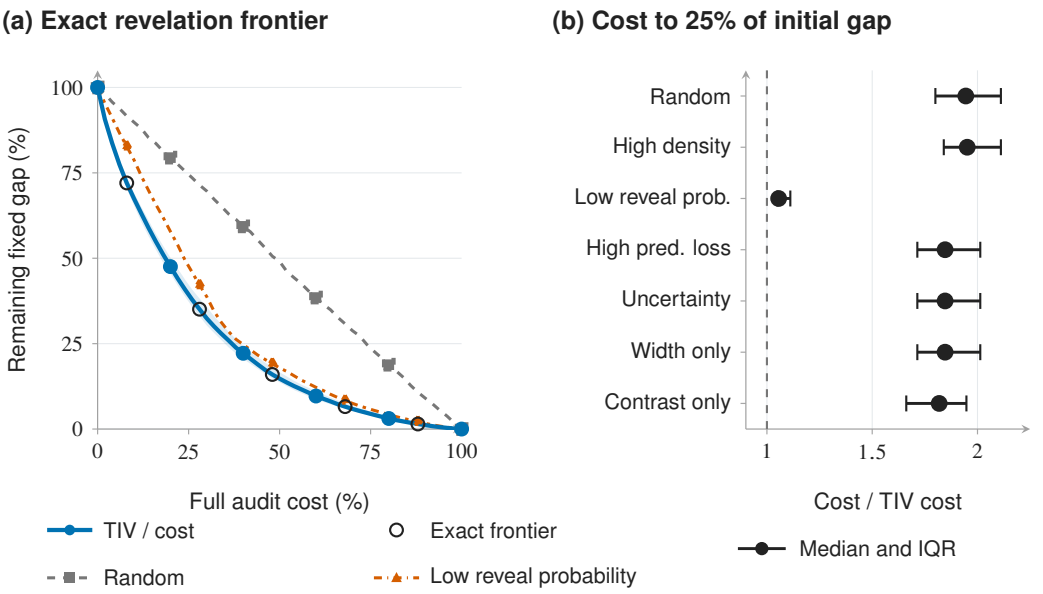


Figure 8: **The exact revelation frontier and the cost of alternative policies.** Across 120 finite-cell worlds, (a) median remaining fixed-threshold ambiguity, normalized within each world; blue shading is the TIV interquartile range and open markers identify the coincident exact frontier. (b) All seven baselines: median cost of reaching 25% of the initial gap relative to TIV in the same world, with interquartile bars. Costs use the recorded 2%-spaced grid.

D.6 Fixed and sequential audits of known repayments

Both protocols use boosting, $\alpha = .95$, $\Gamma = 3$, the 20 controlled masks, and unit cost per audited label. Let $N^{\text{fix}}_{\text{hidden,initial}}$ count initially hidden cases in the fixed protocol's original pool, and let $N^{\text{seq}}_{\text{hidden,initial}}$ count initially hidden cases in the sequential protocol's 80% training pool. Each protocol acquires eight batches of $\lceil .02 N^{\text{protocol}}_{\text{hidden,initial}} \rceil$ labels, capped by the remaining eligible pool. Its budget fraction is the cumulative acquired count divided by that same initial hidden count. Among candidates already known to be hidden, operational TIV is $\widehat{\tau}/(1-\widehat{\pi}) = \widehat{s}_\Gamma |d_{f,\eta}|/\beta$, corresponding to $c = 1 - \pi$ in Appendix C.5.

Protocol	Fixed objective	Sequential retraining
Data and predictor	Original hidden pool; initial cross-fitted predictor and historical nuisance predictions held fixed; upper threshold minimizes the empirical upper-endpoint objective in (6), built from the score (5), before auditing.	80/20 split jointly stratified by (S, Y); acquire only in training pool, refit predictor on all known training labels each round
Evaluation	Empirical mean of remaining-hidden indicators times original per-label TIV; normalize by round zero within each mask	Training-pool retained plug-in width; held-out mean Brier and CVaR on the same reserved 20%
Ordering/ties	One stable decreasing-score ordering per policy; ties retain masked-data order	Partial partition (`argpartition`); no stable secondary batch-cutoff rule; random scores refreshed each round

The fixed protocol takes its upper threshold from the empirical upper-endpoint objective in (6), built from the score (5), using the initial masking predictions of Appendix D.4 and the breakpoint/tie rule of Appendix D.3. It freezes those predictor and nuisance predictions, as well as this threshold, for every acquisition and fixed-gap calculation.

Retained objective. Historical p_1 is fitted on initially revealed training outcomes, clipped as in Appendix D.3, and frozen. For training size n, let $A_{it} \le 1 - S_i$ indicate audit by round t, f_t be the refitted predictor, and $\widehat{p}_i$ the frozen fit. The retained plug-in objective is

$$\begin{aligned}\widehat{Q}_t^{\sigma}(\eta) = \eta + \frac{1}{n\beta}\sum_{i=1}^{n}\Big[S_i a_{f_t,\eta}^{Y_i}(X_i) + A_{it} a_{f_t,\eta}^{Y_i}(X_i) \\ + (1 - S_i - A_{it}) b_{f_t,\eta}^{\sigma}(X_i;\widehat{p}_i)\Big], \qquad \sigma \in \{L, U\}.\end{aligned} \tag{D.3}$$

Known outcomes enter directly; still-hidden outcomes retain historical sensitivity envelopes. Each endpoint uses breakpoint minimization, the threshold tie rule, clipping, and midpoint collapse of Appendix D.3. At full revelation $A_{it} = 1 - S_i$, both objectives equal $\eta + (n\beta)^{-1}\sum_i a_{f_t,\eta}^{Y_i}(X_i)$ and both endpoints equal full-training-data empirical CVaR; a regression test verifies this identity. These training-prediction widths and held-out predictive risks are descriptive trajectories, without fold enlargement or adaptive confidence guarantees.

Acquisition rules. Among still-hidden candidates, let $\widehat{p}_{\text{known}}$ denote the fitted outcome probability from currently known labels, and let $\widehat{s}_\Gamma = q_+(\widehat{p}_1) - q_-(\widehat{p}_1)$ be the sensitivity width from the frozen historical outcome fit. In the sequential protocol, let $\widehat{\eta}^U$ be the current upper-endpoint minimizing threshold of (D.3). The six acquisition scores are TIV per label cost $\widehat{s}_\Gamma |d_{f,\widehat{\eta}^U}|/(\beta\kappa)$, with $\kappa = 1$; independent uniform random scores; predictive uncertainty $1 - 2|f - .5|$; negative fitted probability of currently having a known label; predicted Brier loss $(1 - \widehat{p}_{\text{known}})f^2 + \widehat{p}_{\text{known}}(1 - f)^2$; and historical sensitivity width $\widehat{s}_\Gamma$. The fixed protocol uses its frozen initial predictor, threshold, and historical fits. Sequential TIV uses the current predictor and upper threshold with the frozen historical sensitivity bounds. The propensity baseline refits the indicator of a currently known label on all training covariates, with probabilities clipped to $[.02, .98]$. Predicted loss refits the outcome model on currently known outcomes and uses its raw probabilities. These models share the boosting settings above, with their respective targets, available labels, and seeds. Refreshed acquisition fits guide label choice; retained evaluation continues to use the historical outcome fit in (D.3).

Acquisition rule	Fixed-threshold gap reduction	Final median sequential interval width
TIV per label cost	61%	.035
Random	16%	.067
Lowest revelation propensity	9%	.084
Predicted loss	0%	.116
Predictive uncertainty	0%	.116
Sensitivity width	0%	.116

Table 5: Complete fixed and sequential audit comparison at the nominal 16% budget. Fixed reductions are medians of within-mask percentage reductions; estimated sequential widths are medians across the 20 runs. Integer rounding gives actual median budget fractions of 16.013% of the fixed initial hidden pool and 16.018% of the sequential initial hidden training pool.

Acquisition rule	Estimated width	Mean Brier	CVaR
TIV	0.0348	0.1362	0.8541
Random	0.0671	0.1347	0.8682
Paired TIV−Random	−0.0332	+0.0014	−0.0138
Exact predictor reference			
Constant $f \equiv 1/2$	0.0000	0.2500	0.2500

The constant forecast supplies an exact predictor reference.

Table 6: Precision and predictive loss after eight equal-budget audit rounds. Policy rows are medians across 20 masks; the paired row is the median of the within-mask TIV-minus-random differences. Estimated width uses the training pool; mean Brier loss and CVaR use the same reserved 20% split.

In all 20 paired terminal runs, TIV has lower estimated width and held-out CVaR than random, while random has lower mean Brier loss. The median paired differences are $-.0332$ in width, $-.0138$ in CVaR, and $+.0014$ in mean Brier loss (Table 6). The exact constant reference has lower tail loss and higher mean loss than both policy medians. Figure 9 shows how the estimated width and held-out tail risk evolve as each policy changes the training labels and predictor.

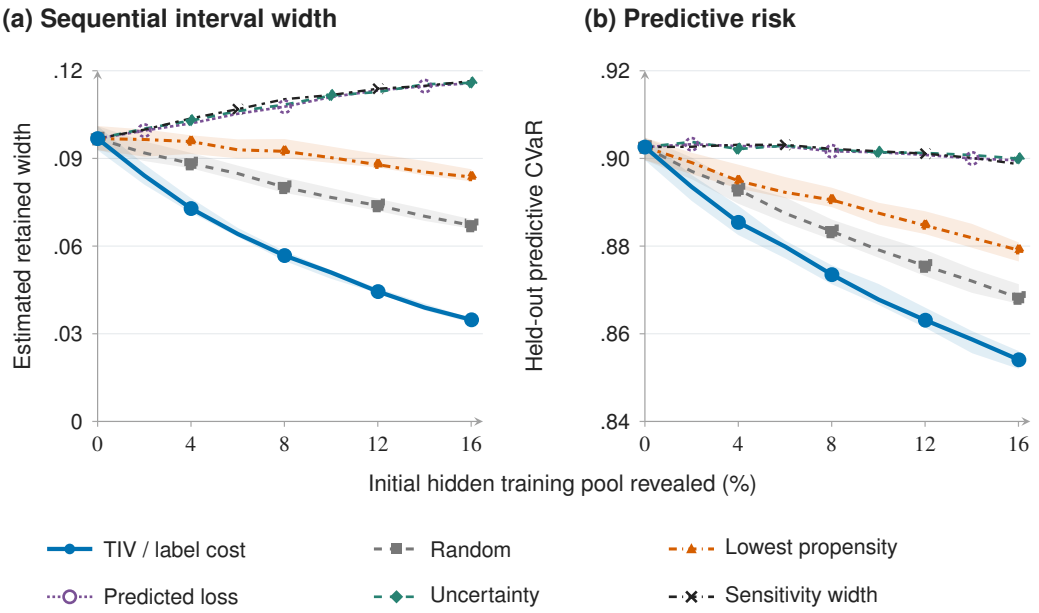


Figure 9: **Sequential precision and predictive outcomes across all six acquisition rules.** Lines are medians over 20 masks at $\alpha = .95$, $\Gamma = 3$; light bands are interquartile ranges for TIV, random, and lowest-propensity acquisition. (a) Estimated retained-sensitivity width for the current predictor. (b) Its CVaR on the reserved 20% split. Predicted-loss, predictive-uncertainty, and sensitivity-width curves are plotted separately and nearly overlap. Lines join recorded rounds. Table 6 gives the exact constant-forecast reference. The horizontal coordinate is the median fraction of initially hidden training-pool labels acquired.

D.7 Recorded implementation inventory

The accompanying `reproducibility/README.md` and `reproducibility/implementation_inventory.tex` record implementation entry points, configurations, seeds, draw order, saved-output mappings, and runtime/hash logs. The sequential results in Figure 4b, Figure 9, and Tables 5–6 use the retained plug-in objective (D.3); the inventory identifies its runner and configuration and records residual-corrected trajectories separately. It also documents the cohort verification: all 53,367 reconstructed issued records match all 12 shared columns in their original order, and all 20 archived mask diagnostics match.

E Sensitivity Calibration and Extensions

E.1 Choosing and calibrating the sensitivity range

For interior probabilities, $\gamma = \log \Gamma$ bounds the hidden–selected log-odds discrepancy:

$$\begin{aligned} |\operatorname{logit} p_0(x) - \operatorname{logit} p_1(x)| &\le \gamma, \\ p_0(x) &\in [\operatorname{expit}\{\operatorname{logit} p_1(x) - \gamma\}, \operatorname{expit}\{\operatorname{logit} p_1(x) + \gamma\}]. \end{aligned} \tag{E.1}$$

Overlap identifies the reference p_1; substantive knowledge or additional outcomes inform the allowed discrepancy, as in sensitivity analysis for nonignorable observation [30–32, 35].

Probability-scale elicitation. For a reference probability $p \in (0,1)$ and elicited hidden range $[l,u] \subset (0,1)$, the smallest symmetric odds bound containing that range is

$$\Gamma_{\text{elic}}(p,[l,u]) = \max\left\{1, \frac{\operatorname{odds}(u)}{\operatorname{odds}(p)}, \frac{\operatorname{odds}(p)}{\operatorname{odds}(l)}\right\}, \qquad \operatorname{odds}(v) = \frac{v}{1-v}. \tag{E.2}$$

For $p = .10$ and $[l,u] = [.05,.20]$, this gives $9/4$. A common Γ must cover the whole target covariate support; uncertainty in p can be included by maximizing over its confidence set.

Pilot calibration and covariate resolution. Declare a finite-cell model $X = g \in \mathcal{G}$, and write $p_{sg} = \Pr(Y = 1 \mid X = g, S = s)$ in cells with originally hidden mass. A pilot identifies p_{0g} only when its design identifies that original hidden distribution: conditional random sampling suffices, while unequal probabilities require adjustment. An arbitrarily selected audited subset instead identifies its own law p_A (Appendix C.1). Suppose confidence sets satisfy $\Pr\{p_{sg} \in C_{sg} \text{ for all relevant } (s,g)\} \ge 1 - \delta_{\text{cal}}$. For sets contained in $(0,1)$, define

$$\Gamma_{\text{cal}} = \exp\left\{\max_{g\in\mathcal{G}} \sup_{\substack{u\in C_{0g} \\ v\in C_{1g}}} |\operatorname{logit} u - \operatorname{logit} v|\right\}. \tag{E.3}$$

On the simultaneous coverage event this bounds every cell's true discrepancy. The supremum can be infinite as admissible probabilities approach incompatible boundaries. Clipping confidence sets to an interior interval preserves coverage only under a justified probability restriction or a confidence construction already supplying it. The finite-Γ restrictions at $p_1 = 0, 1$ are specified in Appendix A.1.

Calibration in (E.3) controls the model with $X = g$. Cell averages do not control discrepancies at finer covariate resolution: either analyze with these cells or justify explicit within-cell homogeneity of the relevant probabilities. A certificate using finer covariates needs control at that finer resolution.

Joint error control. With an independent pilot, condition on it and apply the certificate at the chosen finite Γ_{cal} with valid conditional failure probability δ_{cert}, including nuisance estimation. A union bound gives total failure probability at most $\delta_{\text{cal}} + \delta_{\text{cert}}$ under the two procedures' assumptions. Reusing labels for calibration and revelation requires joint control of the sensitivity choice, audited contribution, and endpoint estimates; their dependence must enter the analysis. For a prespecified finite sensitivity grid, allocate error across its points, each already controlling the stipulated predictor class. A continuum requires uniform control in sensitivity. Let $\Gamma_{\max}$ be the largest sensitivity level under consideration. Population nesting implies that risk-limit or strict-separation claims valid at $\Gamma_{\max}$ hold throughout $[1, \Gamma_{\max}]$. Compare the calibrated range with the breakdown definition in Appendix A.5; Appendix D.3 estimates breakdown numerically by interpolation on a log-sensitivity grid.

E.2 Extensions of the outcome model

Binary outcomes provide two label-specific losses whose ordering supplies one extremal hidden kernel for every threshold. A multiclass extension needs a sensitivity set for the hidden class-probability vector and a new extremality and threshold-optimization argument. Continuous outcomes require conditional-distribution sensitivity sets; unbounded losses additionally require tail conditions for finite statistical control. Population overlap and nuisance-error requirements remain those of Appendix B; the historical extension and masking calibration are specified in Appendices D.3 and D.4.

E.3 Connections to related approaches

Selective-label evaluation concerns outcomes observed through historical decisions [5, 9, 17]. Missing-data analysis and partial identification organize the resulting uncertainty through compatible laws [20–23]; here every point of the tail-risk interval has such a law. The CVaR representation accommodates discrete losses [1, 2, 28, 29]. As in distributionally robust learning [4, 11, 12, 25], the upper endpoint is a worst-case risk; the lower endpoint records the remaining identification information. Orthogonalization and cross-fitting control nuisance bias [7, 14, 27], and Rademacher complexity and contraction control uniform fluctuation [3, 24, 33].

Active learning links queries to a learning objective [8]; TIV links acquisition to the precision of a specified risk claim. Its exact allocation theorem fixes f, η, Γ. The sequential study (Appendix D.6) refreshes acquisition values after retraining while keeping the historical outcome regression fixed in the retained objective. Auxiliary fits to current known-label status or outcomes serve the corresponding acquisition baselines. The study evaluates interval trajectories and held-out predictive risk. Inference across adaptive rounds or at data-dependent stopping requires control of the acquisition history, newly revealed contributions, and stopping rule, beyond the fixed-sample uniform theorem.

F Additional Recorded Experiments

This appendix collects saved experiments that complement the studies in Appendix D: simple observation mechanisms, finite-candidate learning, additional diagnostics for the synthetic revelation frontier, and a finite-population illustration constructed from diabetes data. They answer questions about how the objectives behave under controlled conditions. Their population quantities and empirical trajectories carry no additional finite-sample confidence guarantee. No new simulations or model fits were run when preparing these tables.

F.1 Observation mechanisms, candidate learning, and fixed audits

Low observed risk can hide high population risk. The two-region example has masses $(.82, .18)$, deterministic outcomes $(0, 1)$, and observation probabilities $(.95, .04)$. Use absolute loss and $\alpha = .85$. For the constant-zero predictor, the saved observed-label CVaR is 0.0611, whereas population CVaR is 1.0000. The region-specific predictor $(0, 1)$ has zero loss under both populations. Thus this example demonstrates a large discrepancy between observed and population risk; it does not exhibit a pairwise ranking reversal. The saved envelope calculation uses $\Gamma = 50$ and clips boundary reference probabilities to $[10^{-8}, 1 - 10^{-8}]$, producing negligible nonzero envelope values for the zero-loss reference.

Finite-candidate selection with known auxiliary functions. Use the five-region population specified under "Population sweeps" in Appendix D.2. Outcomes are sampled from its specified Bernoulli probabilities; observation is sampled conditional on the outcome. The adverse-label observation probability in region B is .03, compared with .60 for the other label, giving an odds-ratio sensitivity shift of 48.5. Fix $\Gamma = 50$, $\alpha = .85$, absolute loss, and

$$\mathcal{F}_0 = \{(.05, b, .10, .40, .50) : b \in \{.05, .25, .65, .85\}\}.$$

Each of 250 independent training samples contains 1,400 records. Four rules minimize, over $\mathcal{F}_0$, empirical full-label CVaR, empirical observed-label CVaR, normalized inverse-propensity-weighted observed-label CVaR, or the empirical upper sensitivity objective. The propensity rule uses the known marginal $\pi(X)$; the upper objective uses known p_1 and combines observed excess loss with the bounded hidden contribution. It does not estimate auxiliary models or add the locally robust residual correction. All cutoff searches use exact loss breakpoints. The full-label rule is an oracle benchmark unavailable to a learner with selective labels.

Table 7 evaluates each chosen predictor under the exact population law. Each rule chooses the same candidate in every recorded replication, so the table does not attach a sampling-error bar to its repeated population evaluation. The difference between observed-label and inverse-propensity learning, on one hand, and upper-envelope learning on the other illustrates the consequence of outcome-dependent observation in this particular candidate class. It is not a universal comparison between learning algorithms.

Rule	$f(B)$	Observed CVaR	Full CVaR	Upper endpoint	Width
Observed labels	.25	0.6447	0.7845	0.8877	0.2844
IPW	.25	0.6447	0.7845	0.8877	0.2844
Upper envelope	.65	0.6686	0.6582	0.8628	0.2286
Full-label oracle	.65	0.6686	0.6582	0.8628	0.2286

Table 7: Finite-candidate selection over 250 samples of size 1,400. Values average exact population evaluations of the chosen predictor; each rule selected the same candidate in every replication. The upper endpoint is a population sensitivity envelope, without statistical enlargement.

A single fixed-predictor audit pool. A further sample of 7,000 records from the same population supplies one audit pool. Fix the population observed-label-preferred predictor $f = (.05, .25, .10, .40, .50)$ and the lower-envelope minimizing cutoff $\eta = .5$. Only originally hidden outcomes can be acquired, at unit cost. The five static rankings are per-hidden-label TIV $s_\Gamma |d_{f,\eta}|/\beta$, independent uniform random scores, region mass, uncertainty $-|f - .5|$, and expected absolute loss at the midpoint $(q_- + q_+)/2$ of the sensitivity interval. The implementation sorts each score in decreasing order using NumPy's default `argsort`; it does not impose a separate stable tie rule. The 16

recorded budgets range from zero to 1,800 labels in increments of 120. The predictor and rankings remain fixed throughout.

Revealed outcomes enter the known empirical contribution. The remaining hidden outcomes retain the original sensitivity bounds. Table 8 separately reports the reduction of the fixed-cutoff gap and of the empirical interval whose endpoints minimize their cutoffs independently. Both reductions are normalized by their respective initial value in this one pool. The two objectives need not favor the same allocation at every budget: random revelation removes slightly more optimized width at 600 labels, although TIV removes more fixed-cutoff ambiguity. Prediction uncertainty is also distinct from label-sensitive tail ambiguity: the region with $f = .5$ has identical absolute loss under both labels, hence zero TIV at every cutoff. These are mechanism observations from one realized pool, without replication-based inference or a claim of adaptive learning optimality.

Labels	TIV	Random	Density	Uncertainty	Predicted loss
Fixed-threshold gap removed (%)					
600	42.4	30.6	40.7	15.8	15.8
1,200	76.2	61.6	67.8	47.1	47.1
1,800	100.0	91.4	95.5	88.9	88.9
Separately optimized empirical width removed (%)					
600	29.4	30.3	28.7	21.3	21.3
1,200	72.0	61.0	62.5	59.0	59.0
1,800	100.0	91.1	98.2	87.1	87.1

Table 8: One synthetic pool of 7,000 records, fixed predictor, and unit cost per hidden label. Reductions are normalized by that pool's initial gap or initial optimized width. These two targets need not rank policies identically; at 600 labels random revelation removes slightly more optimized width than TIV.

F.2 Additional diagnostics for the synthetic frontier

Use the same 120 synthetic worlds, each with 120 cells, described in Appendix D.5. The loss is absolute, $\alpha = .85$, and $\Gamma = 12$. The additional saved outputs compare three specified forecasts: p_1, the upper sensitivity probability $q_+(p_1)$, and $p_{\text{full}} = \pi p_1 + (1-\pi)p_0$. The CSV names for these rows are "Observed-selected," "Certificate," and "Oracle." In this experiment they are formulas fixed by the generator, not the results of minimizing empirical risk or an upper endpoint over a candidate class. The full-label forecast uses information unavailable from the observed law. Table 9 reports mean population evaluations over the 120 worlds, alongside the single finite population introduced below. In particular, the $q_+(p_1)$ row is not evidence that the risk-minimizing forecast has been found.

From a common cutoff to separately minimized endpoints. For each world, fix $f = p_1$ and acquire along the TIV-per-cost ordering at $\eta = .5$. Cost is $\sum_j \mu_j c_j \lambda_j$, with a fractional terminal cell; budget fractions divide by the full population cost $\sum_j \mu_j c_j$. The known audited contribution is evaluated with the generative p_0, while the remaining outcomes retain their sensitivity bounds. This isolates the retained population objective from estimation of the newly audited contribution.

The saved diagnostic evaluates 21 cost fractions from zero to one. Its endpoint search uses the grid $\mathcal{G} = \{0, 1/60, \ldots, 1\}$, giving

$$I^{\mathcal{G}}_{\Gamma,\lambda}(f) = \min_{\eta \in \mathcal{G}} U_{f,\eta}(\lambda) - \min_{\eta \in \mathcal{G}} L_{f,\eta}(\lambda).$$

This recorded *grid-minimized width* is distinct from the exact breakpoint-minimized width in Appendix C.4. The companion bound maximizes the common-cutoff gap on the same grid. Because binary absolute-loss contrasts satisfy $|d_{f,\eta}| \le |d_{f,0}|$ and $0 \in \mathcal{G}$, that maximum equals $W_{f,0}(\lambda)$. All 2,520 saved world–budget rows satisfy $0 \le I^{\mathcal{G}}_{\Gamma,\lambda}(f) \le W_{f,0}(\lambda)$ to numerical precision. Table 10 gives selected budgets. This check illustrates the sufficient-width connection; it neither makes the .5-cutoff allocation optimal for optimized width nor turns the finite grid into exact threshold minimization. The gap bound can exceed one because it is not itself a bounded-loss CVaR.

F.3 A diabetes-derived finite-population illustration

Measured outcomes and imposed observation. The construction uses all 442 records in the diabetes dataset distributed with scikit-learn [26]. It binarizes the measured disease-progression outcome at its empirical 75th percentile, assigning $Y = 1$ at or above that cutoff. A standardized logistic regression, fitted to all these outcomes with a 5,000-iteration limit and the recorded software defaults, supplies scores r_i. Fourteen empirical-quantile score cells define the finite covariate population. The score is used to construct strata, not evaluated as a held-out predictive model. In particular, the construction has access to outcomes subsequently called hidden.

For each record, impose the observation probability

$$s_i = \mathrm{clip}_{[.03,.95]}\{.85 - .25r_i - .55Y_i - .20Y_i r_i + \epsilon_i\}, \qquad \epsilon_i \sim N(0, .01^2).$$

The saved construction integrates these probabilities; it does not draw a new binary observation mask. For a cell j with n_j records, set

$$\mu_j = \frac{n_j}{442}, \qquad \pi_j = \frac{1}{n_j}\sum_{i\in j} s_i, \qquad p_{1j} = \frac{\sum_{i\in j} Y_i s_i}{\sum_{i\in j} s_i}, \qquad p_{0j} = \frac{\sum_{i\in j} Y_i(1-s_i)}{\sum_{i\in j}(1-s_i)}.$$

Before evaluation, the implementation clips p_{1j} and p_{0j} to $[10^{-4}, 1-10^{-4}]$ and uses the resulting Bernoulli mixture $p_{\mathrm{full},j} = \pi_j p_{1j} + (1-\pi_j)p_{0j}$. Consequently, the reported full risks refer to this smoothed finite-cell law. The measured progression outcomes and covariates define the construction; the observation channel and acquisition costs are imposed. The population observation share is 0.6325.

Sensitivity, loss, and costs. Use absolute loss and $\alpha = .85$. The sensitivity level is chosen with oracle knowledge of both cell kernels:

$$\Gamma = \max\left\{2,\ 1.10\exp\left(\max_j |\operatorname{logit} p_{0j} - \operatorname{logit} p_{1j}|\right)\right\} = 72.08 \quad \text{(rounded)}.$$

This construction ensures compatibility for the known cell law. It is not a calibration procedure justified using observed labels alone, and it supplies no simultaneous or data-dependent-Γ confidence statement. With zero-based cell indices, the imposed population cost density is $c_j = .65 + 1.15\bar{r}_j + .20\mathbf{1}\{j \bmod 3 = 0\}$, where $\bar{r}_j$ is the cell's mean fitted score.

The three reference forecasts and their population risk evaluations follow Appendix F.2. For the acquisition comparison fix $f = p_1$ and $\eta = .5$, use divisible cell allocations, and evaluate 51 fractions of full cost in increments of .02. The acquisition rules are the seven baselines and TIV-per-cost/frontier ordering described in Appendix D.5. Random acquisition averages the remaining gap over 100 independent uniform-random cell orderings. The 21-budget width diagnostic uses the same 61-point cutoff grid as the synthetic study.

Specified forecast	Observed CVaR	Full CVaR	Upper endpoint
Synthetic populations: mean over 120 worlds			
Selected-label probability p_1	0.8052	0.8703	0.8856
Sensitivity upper probability $q_+(p_1)$	0.8250	0.8102	0.8198
Full-label probability p_{full}	0.7972	0.7464	0.8287
Diabetes-derived finite population			
Selected-label probability p_1	0.2665	0.9252	0.9320
Sensitivity upper probability $q_+(p_1)$	0.8710	0.8693	0.8732
Full-label probability p_{full}	0.5316	0.6760	0.7505

Table 9: Three specified reference forecasts, evaluated with absolute loss and $\alpha = .85$. The CSV role names "Observed-selected," "Certificate," and "Oracle" identify these formulas; they do not denote optimization over a candidate class. Synthetic entries use $\Gamma = 12$; diabetes entries use its oracle-chosen sensitivity level.

Full-cost budget	Synthetic median Grid width	$W_{f,0}$	Diabetes population Grid width	$W_{f,0}$
0%	0.2562	1.3309	0.7295	1.1502
25%	0.0197	0.5393	0.2057	0.5857
50%	0.0080	0.1989	0.0084	0.1484
75%	0.0024	0.0599	0.0002	0.0025
100%	0.0000	0.0000	0.0000	0.0000

Table 10: Width connection along the fixed-$\eta = .5$ TIV-per-cost allocation. Each endpoint is minimized on the recorded 61-point cutoff grid. Synthetic columns are separate medians across 120 worlds; diabetes has one finite population. The sufficient bound is $W_{f,0}$, which is not itself a CVaR risk and can exceed one.

Table 11 reports the first recorded cost fraction at which each curve reaches 25% of its initial fixed-cutoff gap. For random acquisition, this is the crossing of the mean gap curve, rather than the mean of 100 individual crossing costs. TIV and the exact fractional-knapsack frontier coincide by their common ordering. The data illustrate how observation and costs shape a retained population certificate when all relevant quantities are known; they do not establish clinical effectiveness, generalization to a new patient population, or finite-sample coverage. They also do not convert passive revelation into identification of outcomes under untaken actions.

Allocation rule	First recorded cost to 25% gap
TIV/cost	42%
Fractional-knapsack frontier	42%
Random	76%
High density	76%
Lowest observation probability	54%
Predicted absolute loss	44%
Uncertainty	44%
Sensitivity width	44%
Excess-loss contrast	86%

Table 11: Diabetes-derived population: first hit on the 2%-spaced full-cost grid. Random is the mean remaining gap over 100 random cell orderings, so its entry is the first crossing of that mean curve, not the mean of 100 crossing costs. Acquisition is divisible at the cell level.

F.4 Saved-output provenance and reproduction order

The numerical sources are the `experiment1`, `experiment2`, and `experiment3` CSVs, and the `experiment6` and `experiment7_semisynthetic` CSVs under `experiments/results/`. The mechanism examples use the full-sequence random stream with master seed 20260503. The current frontier and diabetes CSVs instead match `run_summary_main_figure.json`, generated by running `experiments/run_tail_observability_experiments.py` with `-main-figure`. The full-run summary's older frontier output uses a different random-stream position. The archived `run_summary_semisynthetic.json` instead records an earlier 569-record malignancy example, with malignancy-rate fields; it is not the 442-record diabetes construction in this appendix. Neither archived output is substituted for the current CSVs. Reproductions should use separate output copies or follow the documented run order so that one entry point does not silently overwrite another run's saved results.

The standard-library extraction script `figures/scripts/prepare_additional_experiments.py` regenerates these tables from the saved CSVs. Its `-check` mode verifies the generated tables, source checksums, agreement with the matching run summary, repeated-sample aggregation, TIV/frontier equality, and the recorded width inequalities. `figures/supplement_additional_provenance.json` records those checks and the source-to-entry-point mapping. The corrected lending sequential experiment, original-cohort verification, and finite-sample endpoint study remain documented in Appendix D and the accompanying reproduction material.

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